ABSTRACT

Predicting NCAA March Madness Outcomes

The NCAA Division I basketball tournament presents a unique prediction challenge due to its single elimination structure and high game to game variability. Three predictive frameworks are developed and evaluated on the Kaggle March Machine Learning Mania competition: a logistic regression model built on Dean Oliver's Four Factors, an XGBoost ensemble incorporating seeding, Elo ratings, and strength of schedule adjusted statistics, and a Linear Regression Monte Carlo model that derives win probabilities from simulated score distributions. All models are evaluated on the 2025 tournament using Brier Score, classification accuracy, and AUC.

XGBoost achieves the strongest probability calibration with a Brier Score of 0.12761 and AUC of 0.9215, while the Monte Carlo model achieves the highest combined accuracy of 79.85% and a comparable Brier Score of 0.12857. The Monte Carlo model was submitted to the 2026 live competition with probability overrides applied to top seeds in early rounds, finishing 822nd with a Brier Score of 0.13237. The base model without overrides finished 387th with a Brier Score of 0.12797.

Jonathen Marin
May 2026

Predicting NCAA March Madness Outcomes

by

Jonathen Marin

A project
submitted in partial
fulfillment of the requirements for the degree of
Master of Science in Mathematics
in the College of Science and Mathematics
California State University, Fresno

May 2026

APPROVED

For the Department of Mathematics:

We, the undersigned, certify that the project of the following student meets the required standards of scholarship, format, and style of the university and the student's graduate degree program for the awarding of the master's degree.

Jonathen Marin

Project Author

Earvin Balderama (Chair)	Mathematics
Steve Chung	Mathematics
Mario Bañuelos	Mathematics

ACKNOWLEDGMENTS

I would like to express my gratitude to Dr. Balderama for his insights and guidance, without which this project would not have been possible. The meetings discussing the basketball landscape and strategies behind March Madness were both enlightening and enjoyable. I would also like to thank my committee members, Dr. Bañuelos and Dr. Chung, whose feedback and advice were invaluable in shaping the final work. I would also like to thank my colleagues whose support and collaboration throughout my coursework made this journey all the more rewarding and possible.

This project would not have been possible without my family and my partner Holly, whose colorful and bubbly energy kept me afloat during the project and throughout the program as a whole. I would also like to thank her little sister Halen, whose wonder for the world was a constant reminder to stay curious. The warmth and laughter from both of them was the fuel that kept me going. Lastly, I would like to thank my father, who worked tirelessly long hours to take care of me, showing me firsthand the true meaning of hard work and a strong work ethic.

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

INTRODUCTION

The National Collegiate Athletic Association (NCAA) serves as the premier developmental system for the National Basketball Association (NBA) and the Women's National Basketball Association (WNBA). With thousands of athletes across both divisions arriving from around the world, the influence of the NCAA is extensive. As the basketball world has evolved into a multi-billion dollar industry [2], the value of a competitive edge has become substantial. Teams historically relied on live scouting and film study to gain these advantages. However, the last decade has seen a major shift toward statistical findings. Inspired largely by the movie *Moneyball*, teams now look to analytics to uncover value that the naked eye might miss.

The focal point of this competitive landscape is March Madness. At the conclusion of the regular season, the selection committee invites 68 teams to participate in this tournament. The field includes the champions from all 32 Division I conferences alongside the best remaining teams chosen at large. The committee seeds these participants into four distinct regions with rankings ranging from 1 to 16. The four teams judged to be the strongest are awarded the number 1 seeds while the next tier receives the number 2 seeds. Within each region, the teams compete in a single elimination format where matchups are initially determined by rank. For example, the top seed plays the sixteenth seed. The winner of each region advances to the Final Four to compete for the national championship. Crucially, these contests take place at neutral sites rather than on home courts, eliminating one of the most reliable advantages in sports. Combined with the single elimination format, where even the strongest teams can be eliminated by a single off night, this structure creates an environment where volatility, the tendency for outcomes to deviate sharply from expectations, is the norm.

The NCAA Men's Basketball Tournament is defined by its structural unpredictability. The single elimination format means that even the most statistically dominant teams face elimination from a single off night, and the compressed six-round schedule amplifies the role of momentum, fatigue, and matchup advantages that aggregate metrics cannot fully capture. This unpredictability is well documented in existing literature. Extensive benchmarking of advanced ranking systems reveals that even the top-performing predictive models hit a ceiling of approximately 75% accuracy, indicating that roughly one in four tournament games ultimately ends in an upset [4]. The consequences for bracket prediction are severe. The difficulty of bracket prediction further highlights this unpredictability. Mathematical projections place the odds of a perfect 63-game bracket anywhere from 1 in 9.2 quintillion for purely random guesses to 1 in 128 billion for educated predictions. Consequently, there is no record of a publicly verified bracket remaining completely intact beyond the Sweet Sixteen [14].

This inherent volatility drives a massive secondary market focused on sports betting. As legal wagering expands globally, March Madness has evolved into one of the largest gambling events in the world, with $3.1 billion wagered on tournament outcomes in 2025 [7]. To manage this risk, bookmakers generate odds that serve as the industry standard for predictive accuracy. These odds represent a market consensus that attempts to price in every available variable from player injuries to historical trends. However, research demonstrates that these betting lines are not strictly driven by probability. Evidence indicates that sportsbook forecasts frequently overreact to new information, creating systematic inefficiencies that violate weak-form market efficiency. This suggests that line-setting is heavily influenced by bookmaker profit margins and public betting sentiment rather than pure statistical likelihood [36]. Consequently, developing a model capable of outperforming these market-established baselines represents the gold standard for predictive success.

This work utilizes the Kaggle March Machine Learning Mania competition as its primary experimental framework. Kaggle provides a setting in which participants submit predicted win probabilities for every possible matchup, rather than making binary bracket selections. The competition evaluates all submissions using the Brier Score, a proper scoring rule that measures the mean squared error between predicted probabilities and binary outcomes [5]. Formally, the Brier Score is defined as:

$$BS = \frac{1}{N}\sum_{t=1}^{N}(f_t - o_t)^2, \tag{1}$$

where N is the number of matchups, f_t is the predicted win probability, and $o_t \in \{0, 1\}$ is the actual outcome. A Brier Score of 0 reflects perfect calibration and a score of 0.25 corresponds to the naive baseline of predicting

Predicted	**Outcome**	**Scenario**	**Brier Score**
0.50	Win or Loss	Baseline	0.2500
1.00	Win	Perfectly confident, correct	0.0000
1.00	Loss	Perfectly confident, incorrect	1.0000
0.70	Win	Moderately confident, correct	0.0900
0.70	Loss	Moderately confident, incorrect	0.4900
0.90	Win	Highly confident, correct	0.0100
0.90	Loss	Highly confident, incorrect	0.8100

Table 1. Brier Score Examples

0.5 for every game. Table 1 illustrates how the score behaves across a range of prediction scenarios.

The table highlights an important property of the Brier Score: overconfident incorrect predictions are penalized more severely than moderate ones. Predicting a 90% win probability for a team that ultimately loses yields a score of 0.81, far worse than the 0.25 baseline. This property incentivizes well calibrated probability estimates rather than extreme predictions, and directly aligns with the objective of this research. The primary goal of this paper is not to compare models against one another but to develop the most accurately calibrated probability estimates possible for submission to the Kaggle competition. Each model is evaluated and refined with this singular objective in mind, and performance is measured exclusively by Brier Score against actual tournament outcomes.

This focus on probability calibration is well suited to basketball because the sport generates a rich and consistent statistical record across all levels of competition. Both individual and collective performances are captured through a standardized set of metrics. Discrete variables such as points, steals, assists, and rebounds provide a consistent numerical foundation for analysis across different eras and levels of competition. This uniformity allows for the creation of structured datasets where team strength can be quantified and compared through objective performance indicators. By leveraging these patterns, machine learning models can identify underlying trends that govern game outcomes and team efficiency.

A basic assessment often relies on a record of wins and losses to infer the overall quality of a team. While these records serve as strong indicators, they frequently fail to capture critical nuance. Furthermore, the single elimination structure of the tournament ensures that upsets remain a constant possibility regardless of historical dominance. A defining example occurred in 2018 when the top ranked Virginia Cavaliers entered the postseason with 31 wins and only 3 losses. They fell to the lowest ranked UMBC Retrievers by 20 points. Such outcomes demonstrate why simple heuristics fail in this environment and why calibrated probability estimates are more valuable than confident binary predictions. Building models capable of mathematically capturing this inherent volatility is the central hurdle for tournament analysts.

Motivated by this challenge, substantial research has addressed the complexities of March Madness prediction. The majority of this work focuses on engineering feature sets for learning algorithms ranging from linear regression to more complex ensemble methods. While the Kaggle competition provides foundational game statistics, transforming these into effective predictors requires significant effort. The following section details the specific data types provided by the competition to establish this baseline. The subsequent section reviews past strategies and the theory behind the modeling techniques. Finally, the paper details the implementation of three distinct modeling techniques, specifically logistic regression, XGBoost, and a Linear Regression Monte Carlo simulation, each evaluated exclusively by their Brier Score performance against the 2025 tournament outcomes, with the best performing model deployed for the 2026 Kaggle submission.

DATA

The highly quantifiable nature of basketball ensures that statistical data is both abundant and accessible. While the Kaggle competition permits participants to incorporate any external datasets they acquire, the organizers themselves provide a comprehensive archive of 37 distinct tables spanning more than three decades, with all tables separated by gender. The archive is divided into regular season and postseason files, with the postseason referred to throughout this project as the playoffs. This research focuses primarily on the detailed results tables for both the regular season and the playoffs, while also utilizing the tournament seeding tables. Although the broader archive extends further back in time, the detailed game statistics required for this analysis commence in the 2003 season, as earlier seasons contain only compact results without the possession-level box score data needed for feature construction. Each entry in the detailed results tables provides a comprehensive statistical breakdown for both the winning and losing teams in every recorded matchup. Table 2 summarizes the specific variables recorded for each team.

While raw box scores capture the foundational events of a game, they often fail to account for nuances such as pace of play and strength of schedule. To address this limitation, this research supplements the Kaggle datasets with advanced metrics from analyst Ken Pomeroy (KenPom), which is incorporated into the Linear Regression Monte carlo model for the men's division. Pomeroy, who is widely accepted as the industry standard for college basketball analysis, normalizes performance by calculating efficiency on a per

Variable	Description
PTS	Points scored
FGM / FGA	Field goals made and attempted
FGM3 / FGA3	Three-point field goals made and attempted
FTM / FTA	Free throws made and attempted
OR / DR	Offensive and defensive rebounds secured
Ast	Assists
TO	Turnovers
Blk	Blocks
PF	Personal fouls

Table 2. Regular Season and Playoff Detailed Result Variables

possession basis, where a possession is defined as the period in which a team controls the ball, ending with a shot attempt, turnover, or score [30]. These ratings are updated daily throughout the regular season and tournament. For the purposes of this study, a snapshot of the rankings was acquired from the day prior to the start of the 2025 tournament. Table 3 describes the primary metrics utilized from this source.

Metric	Description
NetRtg	Differential between Offensive and Defensive Rating
ORtg	Points scored per 100 possessions, adjusted for opponent
DRtg	Points allowed per 100 possessions, adjusted for opponent
AdjT	Number of possessions per 40 minutes

Table 3. KenPom Metrics

Because Pomeroy's ratings are exclusively available for the men's division, this research incorporates an additional dataset from Bart Torvik for the women's division. Torvik cites Pomeroy as a significant influence, and the ratings are constructed using a comparable methodology. However, distinct algorithmic differences exist between the two systems; for further detail regarding these distinctions, the reader is referred to [38]. Table 4 describes the metrics utilized from this source.

Metric	**Description**
ADJOE	Adjusted offensive efficiency
ADJDE	Adjusted defensive efficiency
AdjT	Adjusted tempo

Table 4. Bart Torvik Metrics (Women's Division)

One important data consideration is the 2020 NCAA tournament, which was canceled due to the COVID-19 pandemic. While regular season data exists for the 2020 season, no tournament outcomes were recorded, creating a gap in the historical playoff results that spans both the men's and women's divisions. This means the 2020 season contributes no tournament training observations to any of the models in this analysis, as there are simply no playoff target variables available for that year. Models that train on regular season data may still incorporate 2020 regular season results, but no model draws on 2020 tournament outcomes since none exist.

Across all data sources utilized in this analysis, including the Kaggle box score files, KenPom efficiency ratings, and Bart Torvik rankings, no missing values were encountered that required imputation or row removal prior to modeling. Every team in the dataset had a complete statistical profile across all variables required for feature construction. Any safeguards implemented within individual model pipelines to filter incomplete observations were precautionary measures rather than responses to observed data quality issues. The completeness of the dataset across all sources is partly a consequence of the 2003 cutoff, which ensures that every season in the analysis has a full year of regular season results available before the tournament begins. Additionally, season identifiers, team identifiers, and games played counts are administrative columns present in the Kaggle files that carry no predictive signal and are excluded from all model feature matrices across every modeling framework in this analysis.

One significant challenge in integrating KenPom and Bart Torvik data with the Kaggle dataset is the inconsistency in team nomenclature. For instance, a university might be listed as “St. Mary’s” in one dataset and “Saint Mary’s CA” in another. To resolve this, a two stage matching pipeline is applied. In the first stage, team names from the external source are normalized to lowercase and matched exactly against the Kaggle team spellings lookup file. Teams that match exactly are assigned their Kaggle `TeamID` directly. In the second stage, teams that remain unmatched after exact matching are passed through a fuzzy matching procedure using the Jaro-Winkler string distance metric with a maximum distance threshold of 0.3 [18], where 0 indicates an exact match and 1 indicates no similarity. This threshold was selected through manual inspection of the resulting matches, as values above 0.3 began introducing incorrect pairings while values below 0.3 left valid team name variations unresolved. For each unmatched team the closest spelling in the Kaggle lookup file is selected, allowing minor variations in punctuation, abbreviation, and spacing to be resolved automatically. In practice all teams in both the men’s and women’s divisions were successfully matched, with no teams remaining unmatched.

LITERATURE REVIEW

Past Strategies

The prediction of NCAA tournament outcomes has been studied extensively in both academic and competitive machine learning contexts. Early work demonstrated that tournament seeding alone is a statistically robust predictor of game outcomes, with logistic regression models using a z-score transformation of seed position producing strong win probabilities across historical tournament data [35]. Subsequent work developed a combined logistic regression and Markov chain model that has since been benchmarked against nearly 100 ranking systems, establishing logistic regression as the baseline for tournament prediction [4]. This ceiling has been confirmed empirically, with comparative evaluations finding that no machine learning classifier, including decision trees, random forests, naive Bayes, or multi-layer perceptrons, was able to exceed 74% to 75% accuracy when trained on NCAAB match data. Also, this research established that feature construction matters more than model choice, with adjusted offensive and defensive efficiencies and Dean Oliver's Four Factors consistently ranking as the most predictive attributes regardless of the algorithm applied [41]. These insights have been applied directly to the Kaggle March Machine Learning Mania competition, with early entries constructing ensembles of regression functions trained on statistical differentials such as ranking index differences, field goal percentage, and point differential, showing that fundamental analysis approaches could finish competitively against more complex submissions [13].

The competitive strategies employed by top finishers in the Kaggle competition consistently fall into one of two frameworks that build on this academic foundation. The first relies on fundamental analysis, where the objective is to construct a "true" probability independent of public sentiment. A common implementation involves heuristics such as the Elo rating system, which provides a dynamic measure of team strength that updates after every game and accounts for strength of schedule more effectively than raw records [11]. Many strategies in this framework also incorporate expert efficiency ratings such as KenPom, Nate Silver's FiveThirtyEight ratings, and Massey power ratings, which aggregate performance into a single numeric value. By treating these expert outputs as features rather than final predictions, modelers can leverage domain expertise while correcting for known biases in any individual system.

The second dominant framework relies on the Efficient Market Hypothesis [6]. This approach theorizes that global betting markets, specifically closing lines from major bookmakers, offer a more accurate aggregate predictor of game outcomes than individual statistical models. These strategies do not attempt to beat the market but instead focus on mathematically disentangling the true win probability from the bookmaker's overround and inherent biases. The standard tool in this domain is the `goto_conversion` algorithm [12], which addresses the favorite-long shot bias, the tendency for markets to overvalue underdogs and undervalue heavy favorites, by applying correction functions to raw moneyline prices. By stripping these distortions from closing lines, competitors effectively leverage the crowd wisdom of the market while removing its pricing inefficiencies. For more information on how this strategy works, we refer the reader to [12].

The applicability of these frameworks is heavily constrained by the structural discontinuity introduced by the 2021 Name, Image, and Likeness (NIL) legislation. An analysis of historical team winning percentages in the post-NIL landscape reveals a significant deviation from historical norms, characterized by a rapid consolidation of talent where the number of programs maintaining elite win percentages above 75% has nearly tripled compared to the preceding two decades [3]. This phenomenon implies that feature weights derived before 2021 eras are likely subject to significant concept drift. Consequently, this review restricts its focus to winning strategies from recent Kaggle competitions spanning the last two years, as methodologies developed before the NIL era may not reflect the approaches best suited to the heightened stratification of team strength driven by current transfer portal dynamics.

The most influential baseline in recent competition history is a solution that placed highly in multiple competition years and was directly referenced or built upon by multiple top finishers in both 2024 and 2025 [31]. This solution organizes its feature set into tiers of increasing complexity, beginning with tournament seeding, progressing through season-level efficiency averages, and culminating in a custom Elo rating system and a GLM-based team quality metric derived from point differential. The model is implemented using XGBoost with a fixed number of boosting rounds, and its widespread adoption as a community baseline reflects the consensus that gradient boosting produces strong and reproducible predictions in this competition. A third consistent pattern across recent competition years is the use of simulation to convert win probability estimates into bracket outcomes. Rather than predicting game results once, top competitors simulate each tournament thousands to hundreds of thousands of times, drawing the winner of each game according to the estimated win probabilities. Among the most notable implementations, one approach simulated brackets using win probabilities derived from expert ratings converted through a normal distribution over predicted point margins, while another used opponent adjusted efficiency margins scaled to a standard possession count with a normal distribution and an empirically estimated standard error. This approach of predicting a continuous score or margin first and then simulating to derive a win probability directly inspired the Linear Regression Monte Carlo model developed in this paper.

The review of these strategies provides motivation for the three models developed in this paper. The logistic regression model is grounded in the academic tradition of sports prediction, serving as an interpretable baseline consistent with the feature classes identified as most predictive in the literature [35, 4, 41]. The XGBoost model is built upon the community baseline that dominated both competition years [31], incorporating seeding, strength of schedule adjusted box score statistics, and a custom Elo rating system with cross-validated hyperparameter tuning. The Linear Regression Monte Carlo model draws on the score prediction and simulation pipeline that proved effective among top 2024 finishers, substituting KenPom and Bart

Torvik efficiency ratings for the external rating sources that top competitors relied upon. Together these three frameworks span the primary methodological strategies that have proven most effective in this competition, grounded in the fundamental analysis approach rather than market probabilities, and evaluated exclusively by their Brier Score performance against actual tournament outcomes.

Modeling Techniques

Having established the empirical context and strategic motivation for the selected frameworks, this section provides the formal mathematical definitions for the algorithms that underpin this paper. While the previous section outlined why these models were chosen, the following details how they function mechanically within the prediction pipeline. Specifically, we define the parameter estimation of Logistic and Linear Regression, the probabilistic simulation mechanics of the Monte Carlo method, the sequential tree construction utilized by XGBoost, and cross validation.

Logistic Regression is a binary classification model that estimates the probability of an outcome by applying the logistic (also known as sigmoid) function to a linear combination of input features. The model is defined as:

$$P(Y=1|\mathbf{x}) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\beta_2 x_2+\ldots+\beta_k x_k)}} \tag{2}$$

where Y is the binary outcome, $\mathbf{x} = (x_1, x_2, \ldots, x_k)$ represents the feature vector, and $\boldsymbol{\beta} = (\beta_0, \beta_1 \ \ldots, \beta_n)$ are the model coefficients estimated through the maximum likelihood estimation. This function ensures the predicted probabilities are bounded between $[0, 1]$. Logistic regression assumes a linear relationship between the input features and the log-odds of the outcome. This assumption can be restrictive when the true relationships are nonlinear. To address this limitation, common approaches are feature engineering with polynomial or interaction terms (e.g, $x^2, x_i \cdot x_j$) to capture nonlinear patterns.

Linear regression is a supervised learning method that models the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. Linear Regression is defined as:

$$\hat{Y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_k x_k \tag{3}$$

where $\hat{Y}$ is the predicted continuous outcome, $X = (x_1, x_2, \ldots, x_k)$ represents the vector of features, and $\beta = (\beta_0, \beta_1, \ldots, \beta_k)$ are the model coefficients estimated through ordinary least squares (OLS) minimization. For more information on OLS we refer the reader to [25]. Unlike logistic regression, linear regression produces unbounded predictions and does not inherently output probabilities. In sports prediction contexts, linear regression is commonly used to estimate point totals, scoring margins, or other continuous performance metrics that can subsequently be transformed into win probabilities through additional techniques, such as applying a logistic transformation to the predicted margin or feeding the output into a Monte

Carlo simulation to estimate the probability that a given margin corresponds to a win.

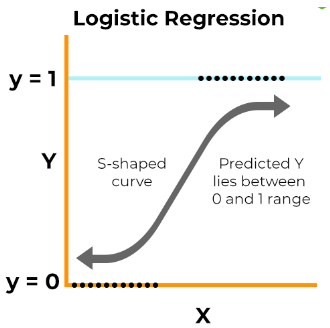

Figure 1. An Example of a Logistic Regression Equation [19].

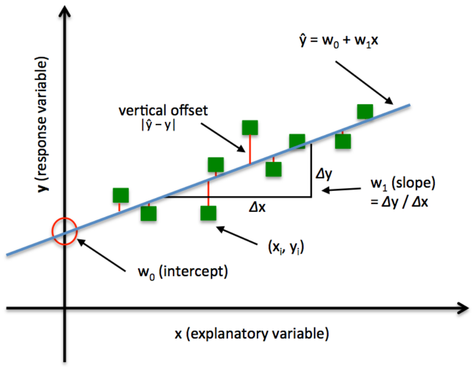

Figure 2. An Example of a Linear Regression Equation [32].

Monte Carlo Simulation is a computational technique that uses repeated random sampling to estimate probabilities for systems with inherent uncertainty. The method generates N independent samples $x_1, x_2, \ldots, x_N$ from a probability distribution $f(x)$, then estimates probabilities by computing the proportion of samples satisfying a certain condition. The simulation is defined as:

$$\hat{P} = \frac{1}{N} \sum_{i=1}^{N} 1(x_i \in A), \tag{4}$$

where $1(\cdot)$ is the indicator function that equals 1 when the condition is satisfied and 0 otherwise. A represents the event of interest, and $\hat{P}$ is the Monte Carlo estimate of the true probability $P(A)$. By the Law of Large Numbers [23], $\hat{P} \to P(A)$ as $N \to \infty$. In sports prediction contexts, Monte Carlo methods are frequently employed to simulate game outcomes by sampling from estimated score distributions, enabling the estimation of win probabilities through empirical frequency counts across simulated matchups.

XGBoost (Extreme Gradient Boosting), considered a "black-box" approach, is an ensemble learning method that constructs a sequence of decision trees where each successive tree is trained to correct the residual errors of the previous trees. Unlike Random Forest, which builds trees independently, XGBoost employs gradient boosting to iteratively minimize a loss function through additive modeling [20]. The model prediction is defined as the sum of predictions from all trees:

$$\hat{y}_i = \sum_{k=1}^{K} f_k(x_i). \tag{5}$$

Here, $f_k(x_i)$ is the prediction of the k^{th} tree for the i^{th} data point, $\hat{y}_i$ is the final predicted value for the i^{ith} data point, and K is the number of trees. Each tree is trained to minimize the objective function

$$\text{obj}(\theta) = \sum_{i}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k) \tag{6}$$

where l is the binary cross-entropy loss function, defined as:

$$l(y_i, \hat{y}_i) = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)], \tag{7}$$

used internally to optimize tree construction. Furthermore Ω is the regularization term, discouraging overly complex trees. The optimization process then proceeds via gradient descent on the residuals. For more information regarding gradient descent we refer the reader to [33]. XGBoost is widely adopted in competitive machine learning for its performance on structured data and ability to capture complex interactions and relationships between features. Figure 3 illustrates how XGBoost combines multiple decision trees sequentially to produce a final prediction.

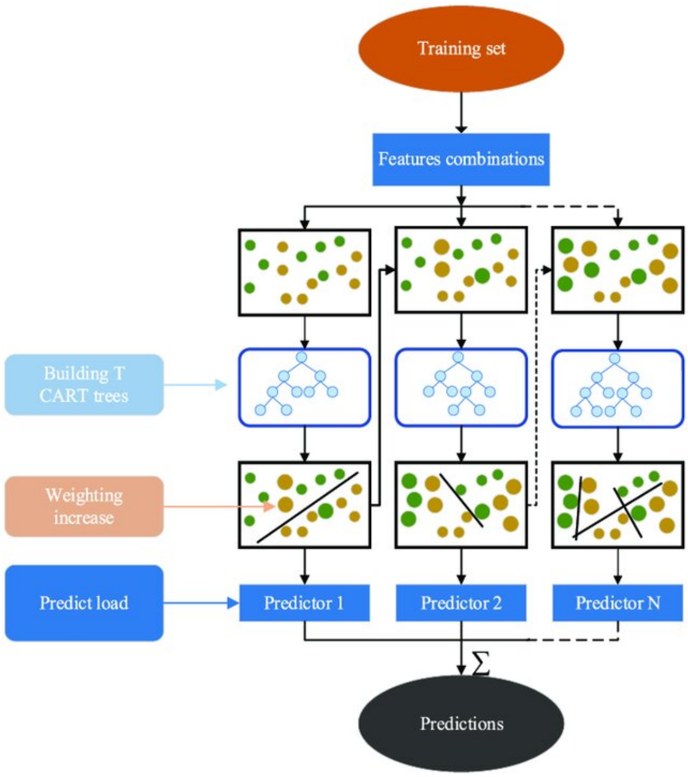

Figure 3. An Illustration Showing How XGBoost Predicts [39].

Cross validation is a model evaluation technique that uses systematic data splitting to assess model performance on data that was not used during training. Rather than evaluating a model on the same data it was trained on, which inherently produces an overly optimistic estimate of performance, cross validation systematically rotates which portion of the data is held out across multiple iterations. This process allows every observation to serve as both a training example and a validation example at different points. In k-fold cross validation, the dataset is partitioned into k equally sized subsets, or folds. The model is trained k times, each time using $k-1$ folds as the training set and the remaining single fold as the validation set [34]. The performance metric is recorded for each fold, and the final cross-validated score is computed as the average across all k folds. This methodology produces a more reliable estimate of generalization performance than a single train-test split, as averaging across multiple held-out subsets reduces dependence on any one particular data partition. For more comprehensive information on cross validation methodology, we refer the reader to [1].

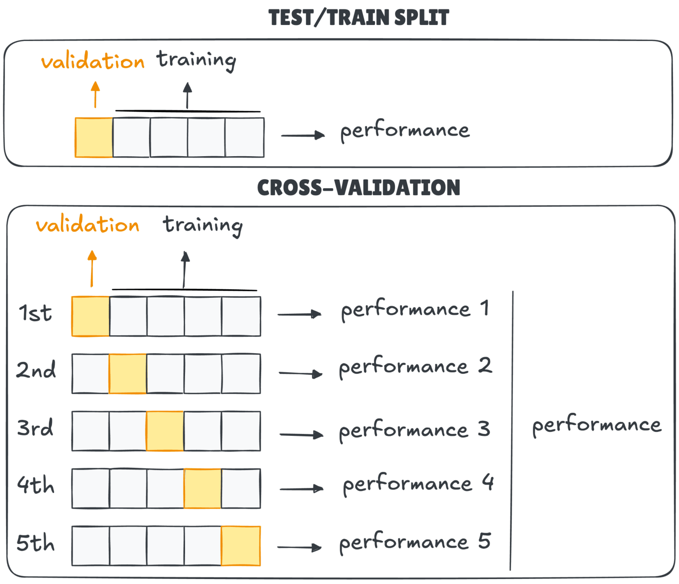

Figure 4. Example of Cross Validation and Regular Test/Train Split [8]

METHODS

The three models in this paper span a deliberate range of complexity and methodology, each grounded in the strategies and techniques established in the preceding literature review. Together they cover the primary approaches identified as most effective in this prediction environment: regression-based probability estimation, gradient boosting on structured efficiency features, and simulating win probability. By testing these three frameworks, this research isolates the specific mechanisms that best mitigate the inherent volatility of single elimination basketball. This design ensures that the theoretical concepts are rigorously validated against multiple analytical philosophies to find the optimal predictive performance.

The simplest approach begins with the premise that a small number of theoretically motivated statistics, computed directly from raw box scores, should be sufficient to capture the fundamental determinants of tournament success. This aligns with the academic finding that feature construction matters more than model complexity, and with the use of logistic regression as the canonical baseline in tournament prediction. The most complex approach abandons that idea entirely, constructing a broad feature matrix and allowing a gradient boosted ensemble to discover relationships across hundreds of variables without prior assumptions about which features matter most, mirroring the community baseline that dominated recent competition years. The third approach occupies a different dimension altogether, bypassing direct classification in favor of predicting each team's expected score and deriving win probabilities through repeated simulation of the resulting score distributions. Each model is presented in its own subsection following a consistent structure covering variable selection, model implementation, and model analysis, This standardization allows for a direct comparison of their respective strengths and predictive capabilities across the tournament dataset. Each model is evaluated against the 2025 tournament outcomes and benchmarked against the winning Brier Score from that competition year to establish its viability.

Logistic Regression

Logistic regression serves as the natural starting point for this analysis due to its direct alignment with the prediction problem. The outcome variable is binary, as a team either wins or loses, and logistic regression models the log-odds of this outcome as a linear combination of input features, producing a calibrated probability as output without requiring additional transformation. Its interpretability is also a key advantage at this stage: the estimated coefficients provide direct insight into which features drive the predicted probability. The independent variables are calculated exclusively from historical regular season data, while the model is trained entirely on historical tournament outcomes. This boundary ensures that the model evaluates how a team's sustained regular season performance translates to the high variance environment of postseason play. To prevent data leakage, the model is trained using a rolling two-year window. For example, to generate forecasts for the 2025 tournament, the model is trained exclusively on tournament matchups from the 2023 and 2024 seasons. This window reflects the structural reality of collegiate athletics, where high roster turnover means a program undergoes a near-complete personnel transformation within two to three seasons, making older data a poor representation of the current team.

Variable Selection

Analyzing raw box score totals is statistically flawed due to the confounding variable of pace. A team that plays at a high tempo may record high totals for points and rebounds simply because they generate more possessions, not because they are inherently more efficient. To address this limitation, the model utilizes the Four Factors of Basketball Success, developed by Dean Oliver [28], which isolate the four components that correlate most strongly with winning: shooting efficiency, turnover avoidance, rebounding dominance, and free throw frequency. Central to these calculations [10] is the estimation of total possessions (Poss), which must be derived from box score data using the following formula. The variable definitions for all abbreviations used in the following formulas are provided in Table 2.

$$Poss = (FGA - OR) + TO + (0.44 \cdot FTA) \tag{8}$$

The first factor is Effective Field Goal Percentage ($eFG\%$), which adjusts for the added value of three-point shots:

$$eFG\% = \frac{FGM + 0.5 \cdot FGM3}{FGA} \tag{9}$$

The second factor is Turnover Percentage ($TOV\%$), which normalizes turnovers against total possessions:

$$TOV\% = \frac{TO}{Poss} \tag{10}$$

The third factor is Offensive Rebound Percentage ($ORB\%$), which measures rebounding dominance relative to available opportunities:

$$ORB\% = \frac{OR}{OR + DR_{opp}} \tag{11}$$

The fourth factor is Free Throw Rate (FTR), which measures how frequently a team converts free throw attempts relative to field goal attempts:

$$FTR = \frac{FTM}{FGA} \tag{12}$$

Beyond Oliver's original framework, the model incorporates Win Percentage ($Win\%$) as a fifth feature, calculated as the mean of binary game results over the regular season. While the Four Factors capture efficiency across specific phases of the game, Win Percentage serves as a composite signal that aggregates the cumulative effect of all factors including those not directly measured, such as coaching decisions, late game execution, and opponent quality. All five features are computed as season-level averages from regular season data and then transformed into differentials between the two competing teams prior to modeling, centering the analysis on the relative mismatch between opponents rather than their absolute statistics.

Model Implementation

The logistic regression model is implemented separately for the men's and women's divisions, with the 2025 model trained on tournament outcomes from the 2023 and 2024 seasons. The decision to train independent models rather than a single combined model with a gender indicator reflects the assumption that the Four Factors carry structurally different predictive weight across divisions. Differences in pace, physicality, and three-point volume suggest that the relationship between efficiency differentials and winning probability is not uniform across genders, and forcing shared parameter estimates would risk masking these structural differences. By isolating the training environments, each algorithm is permitted to optimize its coefficients according to the unique statistical footprint of its respective division.The model uses a logit link function to map the linear combination of input differentials to a probability bounded strictly between 0 and 1. This mathematical transformation is necessary because ordinary linear regression would produce unbounded predictions that cannot be interpreted as viable probabilities. The formal specification for this relationship is defined as:

(13)

$$\text{logit}(\hat{P}) = \ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1(\text{eFG\_diff}) + \beta_2(\text{TOV\_diff}) + \beta_3(\text{ORB\_diff}) + \beta_4(\text{FTR\_diff}) + \beta_5(\text{Win\_diff}),$$

where P represents the probability of a Team 1 victory. The parameter coefficients are derived through maximum likelihood estimation, a process that identifies the $\boldsymbol{\beta}$ vector that maximizes the probability of observing the recorded tournament outcomes given the input differentials. Specifically, the log-likelihood function takes the following form:

$$\sum_{i=1}^{n} \left[y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))\right],$$

where $y_i \in \{0, 1\}$ is the actual observed outcome for the i-th matchup and $p(x_i)$ is the model's predicted probability of a Team 1 victory [24].

The model relies on three core assumptions. First, the log-odds of the outcome are assumed to be a linear function of the input differentials. Second, all tournament game observations are assumed to be independent of one another. Third, the five input differentials are assumed to exhibit no perfect multicollinearity. The first assumption is reasonable given that the features are already constructed as differentials. The second assumption is standard in tournament prediction contexts where each game is treated as an isolated event. The third assumption is verified through the coefficient analysis in the following section.

A practical constraint of this implementation is the relatively small training sample. A single NCAA tournament produces 63 games per division, yielding approximately 126 combined observations per model iteration. While sufficient for a logistic regression with only five features, this sample size meaningfully limits architectural complexity. Overfitting becomes a nontrivial risk if additional interaction terms or polynomial features are introduced. The model is therefore deliberately kept simple, relying exclusively on the five theoretically motivated differentials. This is consistent with the principle of Occam's Razor in model selection [37], where a simpler model that generalizes well is preferred over a more complex one that fits the training data too closely.

Predictions for the 2025 tournament are generated by computing the Four Factors directly from each team's 2025 regular season results. This approach perfectly mirrors the training phase, where every historical tournament game was evaluated strictly against its corresponding regular season efficiency profile. Once compiled, these current metrics are transformed into mathematical differentials for every possible matchup across both the men's and women's divisions. The fitted divisional models are then applied via the inverse logit transformation to produce a calibrated win probability, $\hat{P} \in (0, 1)$, for each candidate pairing. Ultimately, this structural consistency ensures that the final forecasts accurately reflect each team's immediate performance trajectory heading into the postseason, explicitly avoiding the incorporation of outdated statistics from prior academic years.

Model Analysis

With the models estimated on the 2023 and 2024 tournament data, the analysis begins by verifying that the core assumptions of the logistic regression framework are satisfied before interpreting the coefficient estimates. Multicollinearity is first assessed through a Variance Inflation Factor (VIF) analysis for both the men's and women's models. All five predictors fall below a strict VIF threshold of 3 in both divisions, indicating that the input differentials operate as sufficiently independent statistical signals. No single predictor inflates the variance of any other to a degree that would destabilize the coefficient estimates or undermine the reliability of the subsequent significance tests.

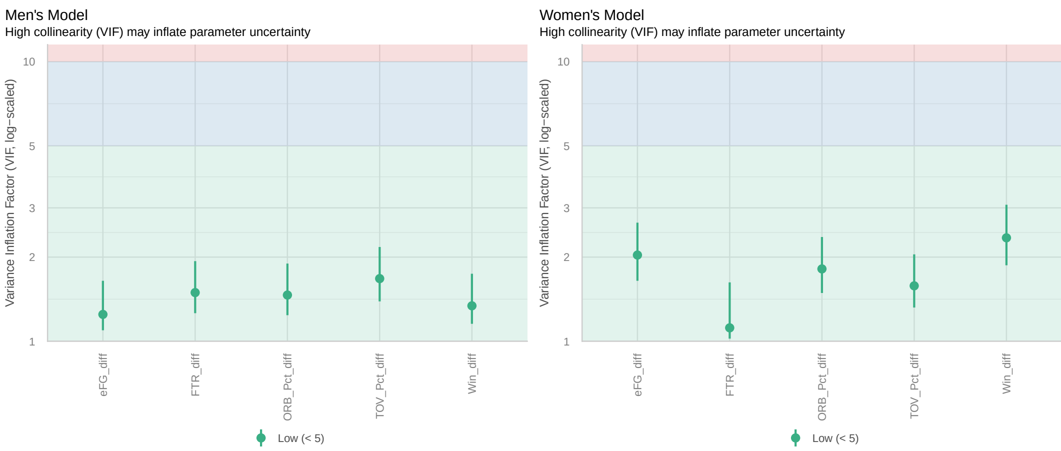

Figure 5. VIF Plots for the Men's and Women's Logistic Regression Models.

The pairwise correlation matrices for the men's and women's training data are presented in Figure 6. In the men's model, all pairwise correlations are modest, with the strongest relationship observed between `TOV_Pct_diff` and `FTR_diff` at 0.53, and between `eFG_diff` and `Win_diff` at 0.39. In the women's model, the strongest relationships are between `eFG_diff` and `Win_diff` at 0.57, and between `TOV_Pct_diff` and `Win_diff` at -0.47. While these moderate correlations are worth noting, none approach a threshold that would indicate problematic multicollinearity, and they are entirely consistent with the VIF results. Together, these two diagnostics confirm that the five differentials can be interpreted individually and that the fundamental collinearity assumption is satisfied in both models.

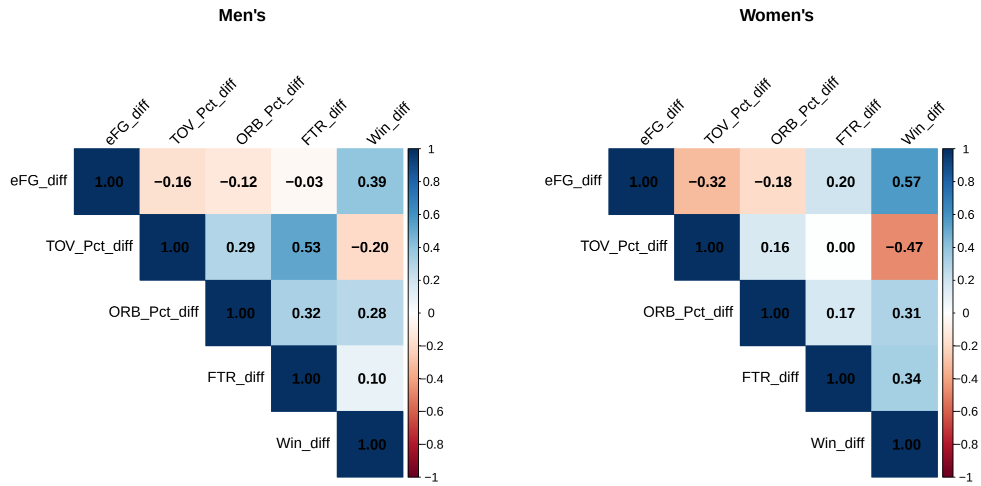

Figure 6. Feature Correlation Matrices for the Men's and Women's Logistic Regression Models.

Because separate models were trained for the men's and women's tournaments, the resulting coefficient structures can be analyzed independently to determine the relative importance of each feature. These independent estimates are then compared against one another to assess whether the Four Factors operate fundamentally differently across the two divisions. By rigorously examining the magnitude and statistical significance of each variable, it becomes possible to identify which specific phases of the game best dictate postseason success for each gender. To illustrate these comparative findings, Tables 5 and 6 present the estimated coefficients, standard errors, p-values, and significance levels for the men's and women's models, respectively.

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	0.4352	0.2062	0.0348	*
eFG Diff	7.7952	6.4160	0.2244	
TOV Pct Diff	-20.5106	12.1028	0.0901	.
ORB Pct Diff	12.3634	4.1219	0.0027	**
FTR Diff	-1.4307	4.5405	0.7527	
Win Diff	1.3253	1.7033	0.4365	

Table 5. Logistic Regression Coefficient Estimates (Men's 2025 Model)

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	-0.0888	0.2055	0.6657	
eFG Diff	17.4458	5.5413	0.0016	**
TOV Pct Diff	-19.8212	8.7655	0.0237	*
ORB Pct Diff	12.4308	3.5127	0.0004	***
FTR Diff	3.9742	5.1691	0.4420	
Win Diff	-1.8468	2.2764	0.4172	

Table 6. Logistic Regression Coefficient Estimates (Women's 2025 Model)

In the men's model, the only variable to reach conventional statistical significance is Offensive Rebound Percentage, with an estimated coefficient of 12.3634 and $p = 0.0004$. The positive sign is consistent with Oliver's framework [28], as a team that dominates the offensive rebounds generates additional possessions while simultaneously denying the opponent transition opportunities. Turnover Percentage approaches significance at the 0.10 level with an estimate of -20.5106, and the negative direction is theoretically correct, as a team that surrenders possessions more frequently than its opponent is expected to win less often. Effective Field Goal Percentage, Free Throw Rate, and Win Percentage all fail to reach significance in the men's model, suggesting that within the narrow two-year training window, shooting efficiency and win records carry limited additional explanatory power once rebounding and turnover control are accounted for.

The women's model tells a markedly different story regarding the underlying drivers of tournament success. Three variables reach statistical significance: Offensive Rebound Percentage ($\hat{\beta} = 12.4308$, $p < 0.001$), Effective Field Goal Percentage ($\hat{\beta} = 17.4458$, $p = 0.0016$), and Turnover Percentage ($\hat{\beta} = -19.8212$, $p = 0.0237$). The near identical magnitude of the Offensive Rebound Percentage estimates across both independently trained models, 12.36 in the men's and 12.43 in the women's, strongly suggests that second chance scoring is a universal determinant of tournament success regardless of division. The most substantive structural difference between the two models is the role of Effective Field Goal Percentage, which is the second most significant predictor in the women's model but statistically indistinguishable from zero in the men's. This divergence suggests that shooting efficiency differentials are more decisive in women's tournament outcomes, where a meaningful gap in $eFG\%$ translates more reliably into a victory than in the men's game, where superior athleticism and individual shot creation can often compensate for overall efficiency deficits. Free Throw Rate fails to reach significance in either model, which is consistent with the relative stability of free throw volume across elite programs at the tournament level.

The significant intercept in the men's model warrants brief discussion. The intercept estimate of 0.4352 represents the log-odds of a Team 1 victory when all five statistical differentials are exactly zero, simulating a matchup between two teams with perfectly identical efficiency profiles. Theoretically, the probability of either team winning in a perfectly symmetric matchup should be exactly 0.5, which corresponds to a log-odds of zero and a statistically insignificant intercept. However, the fact that this intercept is statistically significant at the 0.05 level suggests a slight bias within the training data, where the lower ID team won more frequently even after controlling for all efficiency metrics. Because Team 1 is strictly defined as the team with the numerically lower Kaggle ID, a static designation originally assigned alphabetically rather than by program prestige or talent, this finding clearly does not reflect a genuine basketball phenomenon. Instead, it is most likely a small sample artifact introduced by the restricted training window and should not be interpreted as a meaningful predictive signal for future tournaments.

The discriminative ability of both models is further assessed through the Area Under the Receiver Operating Characteristic Curve (AUC), visually presented in Figure 7. The AUC formally measures the probability that the model assigns a higher predicted probability to a randomly selected win than to a randomly selected loss. Within this statistical framework, a baseline value of 0.5 indicates no predictive ability, equivalent to random guessing, while a value of 1.0 signifies perfect predictive discrimination. The men's model achieves a training AUC of 0.7056, and the women's model achieves a training AUC of 0.7775. These resulting metrics confirm that both individually trained models exhibit meaningful predictive power on the historical data, successfully separating wins from losses well beyond the threshold of random chance.

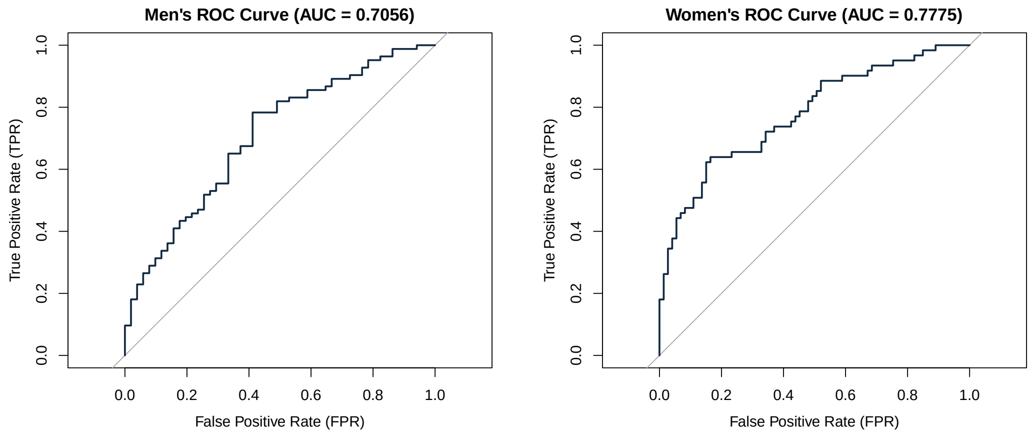

Figure 7. ROC Curves for the Men's and Women's Logistic Regression Models.

Boosting

While logistic regression provides an interpretable and theoretically grounded baseline, its core assumption of a linear relationship between the input differentials and the log-odds of winning imposes meaningful constraints on its predictive ceiling. In reality, the fundamental relationship between team efficiency metrics and tournament success is often highly nonlinear and subject to complex statistical interactions. For instance, a team with a dominant offensive rebounding advantage may only convert that advantage into wins when paired with a sufficient shooting efficiency threshold. This type of conditional relationship is exactly what a linear model cannot capture without explicit feature engineering, prompting the extension of the modeling framework to include XGBoost.

XGBoost is a natural candidate for this problem for several reasons. Unlike traditional parametric models, it does not require strict assumptions about the distributional form of the predictors or their linear relationship to the outcome variable. It is also inherently robust to the scale differences between features, meaning the large magnitude differences observed between metrics in the logistic regression output do not require normalization prior to fitting. Furthermore, XGBoost includes regularization parameters that directly penalize model complexity, which is particularly valuable given the modest total number of tournament observations available across both divisions.

A critical design decision in the data construction phase is the symmetric duplication of each historical tournament game. For every game in the dataset, two distinct rows are created: one where the winning team is assigned to the Team 1 position, and one where the losing team is assigned to the Team 1 position. The binary outcome variable is then strictly set to 1 when Team 1 wins and 0 when Team 1 loses. This duplication serves three distinct purposes: it allows the model to learn from both the winner's and loser's perspective, it ensures the algorithm does not develop a positional bias toward the Team 1 slot, and it guarantees an exactly balanced target distribution of 50% wins and 50% losses.

The final curated training matrix constructed for this ensemble consisted of $4,552$ rows and 139 features, yielding approximately 33 observations per feature. While this ratio would be highly problematic in a traditional parametric setting, XGBoost's rigorous regularization framework and feature subsampling at each tree mitigate this concern by constraining how aggressively the model exploits the feature space. By utilizing these advanced boosting mechanisms, the model can safely navigate a highly dimensional dataset without suffering from immediate overfitting. The following subsections detail the specific variable selection and implementation decisions made for this boosting framework, mirroring the structure of the logistic regression analysis to facilitate a direct comparison between the two approaches.

Variable Selection

A key motivation for the breadth of this feature set is that XGBoost's internal regularization framework absorbs the cost of including additional variables in a way that ordinary regression cannot. In logistic regression, adding correlated or weakly predictive features inflates standard errors, destabilizes coefficient estimates, and risks overfitting, which is why the Four Factors model was deliberately restricted to five predictors. XGBoost does not share this vulnerability. At each boosting round the algorithm selects only the features and split points that produce the greatest reduction in the loss function, effectively ignoring features that contribute no signal. Features that are redundant are simply passed over during tree construction rather than distorting the model as they would in a linear framework. This property is reinforced by the `colsample_bytree` hyperparameter, which randomly samples a subset of features at each tree, further reducing the influence of any single uninformative predictor. As a result, the XGBoost feature set can be expanded aggressively without the same penalty that would apply in a parametric setting, and the cost of including a feature that turns out to be irrelevant is negligible compared to the cost of omitting one that carries genuine signal.

The XGBoost model is trained on all NCAA tournament games between 2003 and 2024, combining both the men's and women's divisions into a single dataset. This differs from the logistic regression approach, where the divisions are modeled separately. The decision to combine divisions here is deliberate: XGBoost is a high-capacity model with a substantially larger feature set, and merging both divisions increases the number of available training observations. Given that the total number of tournament games in any single division across 22 seasons is modest relative to the complexity of the feature space, this pooling strategy reduces the risk of overfitting and allows the model to learn more feature relationships. The underlying assumption is that the predictive structure of tournament outcomes, specifically the relationships between efficiency differentials, seeding, and win probability, is sufficiently consistent across divisions to justify training on a combined sample. To ensure the model can still account for any systemic, structural differences between the two tournaments, a binary `men_women` indicator variable is retained in the feature set. This allows the algorithm to learn universal interactions across the pooled data while applying bracket-specific adjustments when necessary.

The feature set for the XGBoost model is organized into three tiers of increasing complexity: seeding features, simple season statistics, and strength of schedule adjusted box score profiles. Together these tiers provide the model with information about committee evaluation, raw performance, and opponent-adjusted efficiency. The first tier consists of tournament seeding. Each team's seed number is included individually as `T1_seed` and `T2_seed`, along with their difference `Seed_diff`, defined as the opponent's seed minus the team's seed such that a positive value indicates a favorable matchup. Including all three seed features rather than the differential alone allows the model to learn nonlinear effects of absolute seeding. A seed differential of one means something very different when it separates a 1 seed from a 2 seed than when it separates an 8 seed from a 9 seed. The lower seeded teams in the tournament are generally far more dominant over their opponents than the differential alone would suggest, and a model that only sees the difference cannot distinguish between these cases. Figure 8 confirms this through two complementary views. The left panel shows that average point differential declines monotonically as seed number increases across both divisions, while the right panel shows that as the seed differential between two teams widens, the better seeded team wins by larger margins on average, validating seed differential as a strong predictive signal in its own right.

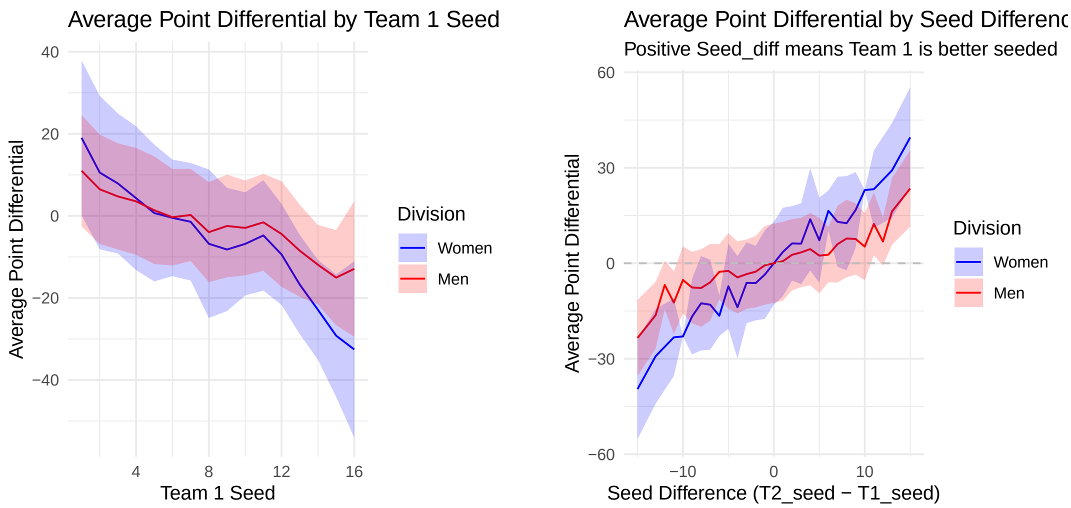

Figure 8. Left: Average point differential by Team 1 seed. Right: Average point differential by seed differential.

The second tier consists of simple season statistics computed from regular season games. For each team these include season win percentage, average points scored, average points allowed, and average point differential. Pairwise differentials between Team 1 and Team 2 are also computed for each of these statistics, providing the model with both absolute team quality signals and direct matchup comparison signals. These features capture overall team strength but do not account for the quality of opposition faced. A team that averaged 85 points per game against a weak conference schedule is not necessarily more dangerous than a team that averaged 75 points against a gauntlet of top 25 opponents, and the raw averages alone cannot make that distinction. This limitation motivates the third tier.

The third tier introduces strength of schedule adjusted box score profiles. For each of the fourteen raw box score statistics available in the Kaggle data, including field goals made and attempted, three point attempts, free throw attempts, offensive and defensive rebounds, assists, turnovers, steals, blocks, and personal fouls, seasonal averages are computed from two perspectives. The first perspective captures a team's own offensive and defensive production, prefixed as `Off_` and `Def_` respectively. The second perspective captures the quality of opposition faced by averaging the seasonal profiles of every opponent a team played during the regular season, producing a set of `SOS_` prefixed features. A team whose offensive statistics were produced against consistently strong defensive opponents is credited more heavily than a team whose statistics were inflated by a weak schedule. All box score statistics are adjusted for overtime prior to averaging using the formula $\texttt{adjOT} = \frac{40+5\times\texttt{NumOT}}{40}$, which scales each game's statistics back to a standard 40 minute equivalent. This ensures that high scoring overtime games do not artificially inflate a team's seasonal averages.

The final component of the feature set is a custom Elo rating system developed to capture dynamic team strength across the full regular season. Unlike the static season averages described in the previous tiers, Elo ratings are updated after every game and reflect the cumulative result of a team's performance relative to its opponents throughout the season. This means a team that starts slowly but peaks heading into the tournament will have a higher Elo rating than its season averages alone would suggest, and conversely a team that fades late in the season will have a deflated rating relative to its win percentage.

The Elo system used here is adapted from the rating system originally developed for chess by Arpad Elo [11], with the implementation structure inspired by [16], originally written in Python and reimplemented in R for this analysis. Each team begins the first season with an initial rating of 1500. After each regular season game the winner gains rating points and the loser loses an equal number, preserving the total rating mass in the system. The magnitude of the exchange depends on how surprising the result was relative to the pre-game expectations implied by the two teams' current ratings. The expected win probability for Team 1 is computed as:

$$E_1 = \frac{1}{1 + 10^{(R_2 - R_1)/400}} \tag{14}$$

where R_1 and R_2 are the current ratings of Team 1 and Team 2 respectively. The denominator of 400 is the width parameter that controls how sensitive the expected probability is to rating differences. After the game resolves, ratings are updated as:

$$R_1 = R_1 + k(1 - E_1)$$
$$R_2 = R_2 + k(0 - E_2)$$

where $k = 64$ is the K-factor controlling the maximum possible rating change per game and $E_2 = 1 - E_1$. When a heavy favorite wins, E_1 is close to 1 and the rating change is small since the result was expected. When a heavy underdog wins, E_1 is close to 0 and the rating change is large. This property makes Elo particularly well suited for tournament prediction because late season upsets and dominant runs cause large rating movements that static averages cannot capture. A key design decision in this implementation is the Elo carry over. Rather than resetting every team to 1500 at the start of each new season, the end of season rating is partially carried forward according to:

$$R_{\texttt{new}} = 0.5 \times R_{\texttt{final}} + 750. \tag{15}$$

This means 50% of a team's historical rating is preserved into the next season while the other 50% regresses toward the mean. The motivation is that program quality is not purely year to year. A program like Duke or Connecticut that has been consistently elite for two decades should not start each season at the same rating as an average program making its first tournament appearance. The carry over allows the Elo system to accumulate this historical signal while the regression toward the mean prevents ratings from drifting arbitrarily far from the initial value over many seasons. Figure 9 illustrates the Elo ratings for the top 20 men's and women's programs in 2025, with the left panel displaying the men's rankings and the right panel displaying the women's rankings, reflecting both current season performance and accumulated historical quality. Each team's final Elo rating is included in the feature set as `T1_Elo` and `T2_Elo`, along with their difference `Elo_diff`.

Including all three allows the model to learn both absolute and relative rating effects simultaneously.

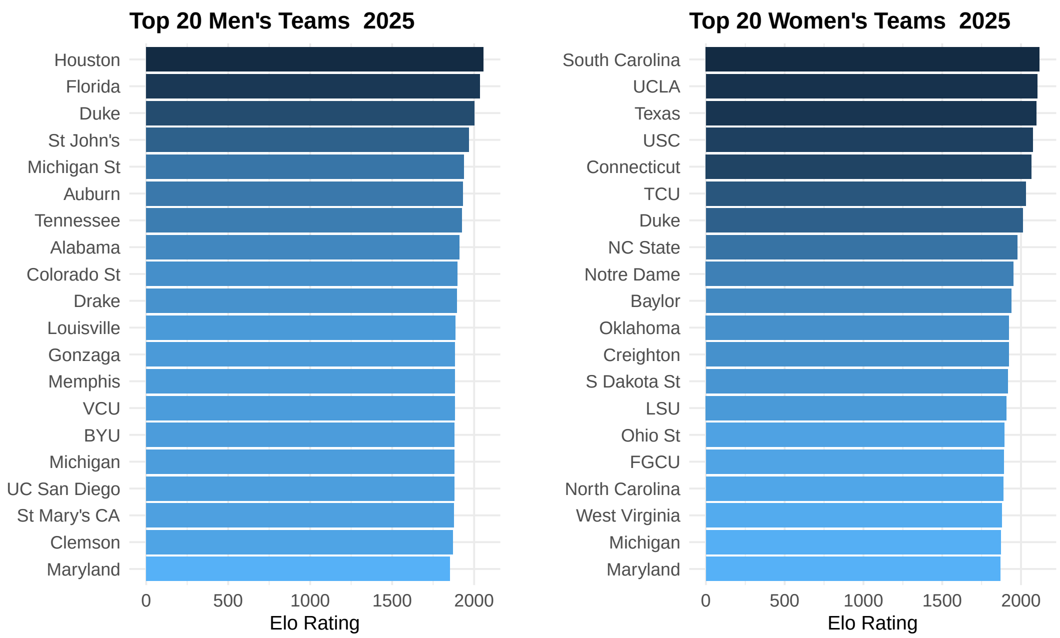

Figure 9. Left: Top 20 men's programs by end of season Elo rating in 2025. Right: Top 20 women's programs by end of season Elo rating in 2025.

Model Implementation

With the training matrix assembled and the feature set finalized, the focus shifts to training the XGBoost model and selecting the configuration that produces the best performance. The model is trained using a binary logistic objective function, which directly optimizes for calibrated win probabilities rather than predicting a continuous point differential that requires a subsequent transformation step. A histogram-based tree method is employed to maximize computational efficiency across the expanded feature space, and a static random seed is set to guarantee reproducibility. Two key decisions must be made before training can begin: the number of boosting rounds and the hyperparameter values that control the complexity and regularization of each individual tree. Neither can be determined from the training data alone without risking overfitting. To address this, five-fold cross validation, as defined in the Modeling Techniques section, is applied to the full tournament training dataset. At each fold, the model is trained on 80% of the tournament games and evaluated on the held out 20%, with the average loss across all five folds used as the selection criterion. Early stopping is applied with a patience of 50 rounds, meaning training halts automatically if the cross validated performance does not improve for 50 consecutive rounds. Cross validation identified 415 as the optimal number of boosting rounds. Five folds was chosen as a balance between bias and variance in the performance estimate, as fewer folds reduce the training set size while more folds increase computational cost and can produce high variance estimates on small datasets.

In addition to the number of boosting rounds, five hyperparameters were tuned to control the complexity and regularization of the model. Each parameter was tuned independently by fixing all others at their current best value and testing candidate values one at a time, selecting the value that produced the lowest cross validated loss before moving to the next parameter. The learning rate `eta = 0.01` was selected after testing values of 0.02 and 0.005, producing a smoother and more regularized model at the cost of requiring more boosting rounds to converge. A value of `max_depth = 5` was selected after testing values of 4 and 6, controlling how many sequential splits each tree can make. The `subsample = 0.8` parameter means each tree is built on a random 80% of the training data, introducing stochasticity that reduces overfitting. Similarly, `colsample_bytree = 0.7` means each tree randomly samples 70% of the available features, further reducing the influence of any single predictor. Finally, `min_child_weight = 4` sets the minimum number of training observations required in a leaf node before a split is made, preventing highly specific splits that apply to only a handful of games. Predictions for the 2025 tournament are generated by constructing the full feature matrix for every possible matchup using each team's 2025 regular season statistics and final Elo ratings, then applying the fitted model to produce a win probability $\hat{P} \in (0, 1)$ for each pairing.

Model Analysis

The feature importance output from the final model ranks each predictor by its average gain across all splits, where gain measures the reduction in the loss function attributable to each specific feature. As shown in Figure 10, `Seed_diff` is the most influential predictor by a wide margin, accounting for 32.2% of the total gain. This result is entirely consistent with the exploratory analysis presented in the Variable Selection section, which demonstrated a remarkably strong relationship between seed differential and tournament outcomes. Ultimately, this confirms that tournament seeding remains the single strongest signal for predicting matchup results across the historical dataset.

The `Elo_diff` metric ranks second with 11.5% of the total gain, capturing cumulative season long performance in a nuanced way that seeds alone cannot reflect. The `avg_point_diff_diff` variable contributes an additional 5.5%, reflecting the average margin by which each team outperformed its opponents during the regular season. Beyond these three primary features, the remaining predictors each contribute less than 3% individually, with strength of schedule variables making up the next tier followed by a long tail of specific box score statistics. This concentration of predictive power in seeding and Elo suggests that broad measures of aggregate team quality drive tournament outcomes significantly more than any individual box score metric.

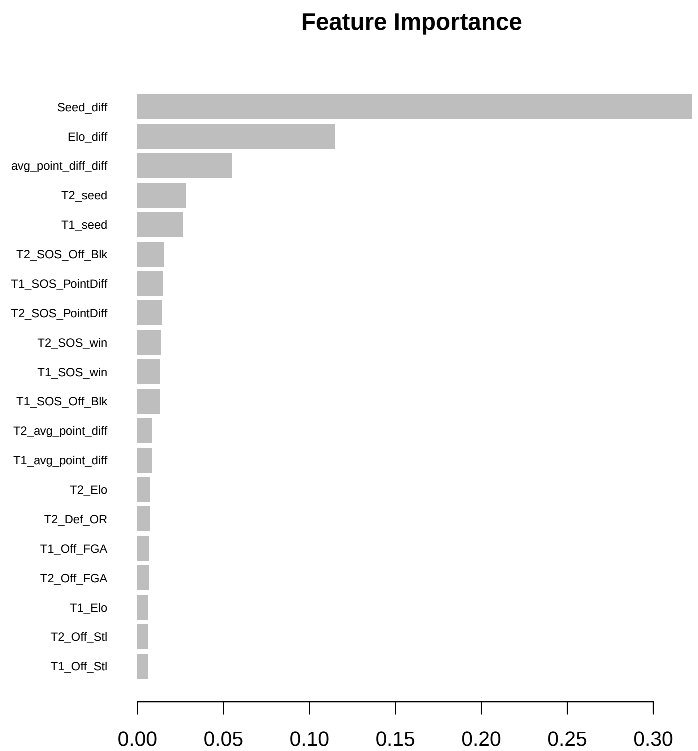

Figure 10. Feature Importance for the XGBoost Model.

The predictive ability of the model is further assessed through the (AUC), presented in Figure 11. The training AUC of 0.9235 confirms that the model possesses strong predicative ability on the training data, meaningfully exceeding the 0.5 baseline that would be expected from a model with no predictive power. This high level of discrimination is consistent with the feature importance findings, where a small number of features like seeding and Elo dominate the model's capacity to separate wins from losses. Because the AUC is significantly elevated, it provides additional confidence in the model's ability to rank team strength effectively prior to the 2026 deployment.

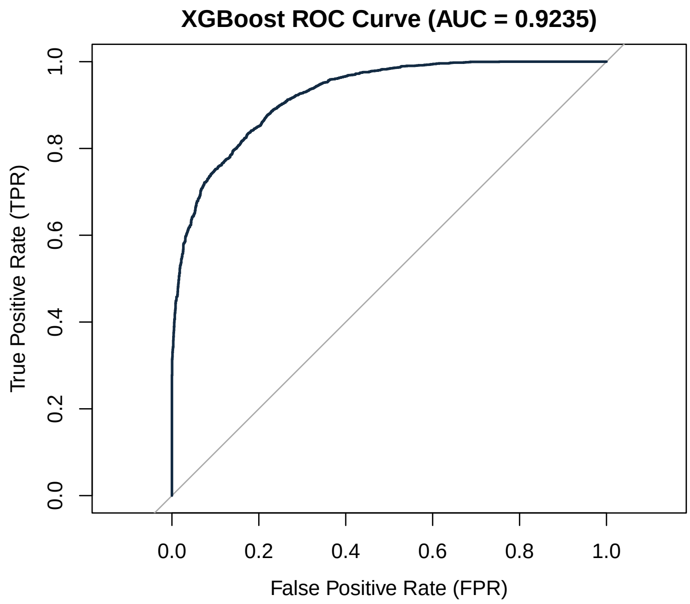

Figure 11. ROC Curve for the XGBoost Model

Linear Regression Monte Carlo

The logistic regression and XGBoost models developed in the preceding sections predict win probabilities directly from features derived entirely from the Kaggle competition dataset. Rather than extending those frameworks further, the Linear Regression Monte Carlo model takes a different approach to the prediction problem. Where the previous two models learn a direct mapping from team statistics to win probability, this model grounds the prediction in an explicit scoring model, asking first how many points each team is expected to score before deriving a win probability from the simulated distribution of outcomes. The prediction pipeline is decomposed into two stages. In the first stage, a linear regression model is trained to predict the number of points a team will score given its own offensive efficiency, its opponent's defensive efficiency, and the pace at which both teams play. In the second stage, the predicted scores and their associated uncertainty are used to simulate each game thousands of times, with the win probability derived as the proportion of simulations in which the team of interest outscores its opponent. This simulation based framework incorporates external efficiency ratings from two sources: KenPom for the men's division and Bart Torvik for the women's division. Both sources provide opponent adjusted offensive and defensive efficiency metrics that are not available in the raw Kaggle box score data, allowing the model to draw on a richer and more carefully constructed measure of team quality than the previous two models

The Linear Regression Monte Carlo model is motivated by the observation that predicting a final score is a more tractable regression problem than directly estimating a win probability. Rather than asking the model to learn a complex nonlinear boundary between wins and losses, the linear regression stage asks a simpler question: given two teams' efficiency profiles, how many points should each team score? The residual standard deviation from this regression, denoted σ, captures the inherent unpredictability of college basketball scoring and serves as the engine of the simulation stage. By drawing repeated samples from the score distributions and counting the proportion of simulated games in which each team wins, the model converts a point estimate into a probability that naturally reflects the uncertainty in the underlying score prediction.

Variable Selection

The feature set for the Linear Regression Monte Carlo model is built around opponent-adjusted efficiency ratings sourced externally rather than computed from raw box score totals. For the men's division, KenPom provides adjusted offensive efficiency (`OffRating`), adjusted defensive efficiency (`DefRating`), and adjusted tempo (`AdjTempo`), all of which account for the strength of opposition faced and the pace at which games were played [30]. For the women's division, equivalent metrics are sourced from Bart Torvik, providing adjusted offensive efficiency (`AdjOE`), adjusted defensive efficiency (`AdjDE`), and adjusted tempo (`AdjT`) [38]. These opponent adjusted metrics are preferred over the raw box score averages available in the Kaggle dataset because they isolate a team's true efficiency from the confounding influence of schedule strength, providing a cleaner signal for predicting how many points a team will score against a given opponent. Unlike the previous two models which draw on historical tournament outcomes for training, the linear regression stage is trained exclusively on the current regular season, as the external efficiency ratings from KenPom and Bart Torvik are season snapshots that reflect the adjusted efficiencies accumulated over that season's play.

A key design decision in the feature construction is how the efficiency ratings are paired for each prediction. Rather than inputting each team's statistics in isolation, the model explicitly pits each team's offensive profile against the opponent's defensive profile. To predict Team A's score, the model uses Team A's adjusted offensive efficiency as `OffRating`, or `AdjOE`, and Team B's adjusted defensive efficiency as `DefRating` or `AdjDE`, along with both teams' tempo values. The same logic applies symmetrically to predict Team B's score. This pairing structure mirrors how scoring actually occurs in basketball, a team's expected output is not simply a function of how well it attacks, but how well it attacks against a specific opponent's defense. A team with an elite offense facing an elite defense should be expected to score fewer points than the same team facing a weak defense, and the feature pairing ensures the model captures this interaction directly rather than leaving it to be inferred from aggregate statistics alone.

Two additional differential features are included to capture relative team quality beyond the efficiency ratings alone. The Elo differential is computed as the difference between each team's Elo rating, utilizing the same custom implementation described in the XGBoost model [16]. For the men's division, an additional net rating differential is also included, defined as the difference between each team's KenPom net rating, which represents the margin between adjusted offensive and defensive efficiency. This feature is not included in the women's model because the Bart Torvik data source does not provide a precomputed net rating, and computing it manually from `AdjOE` and `AdjDE` was not performed during implementation. This represents a known limitation of the women's model relative to the men's, and future iterations should compute and include this feature to fully align the two frameworks. Ultimately, the presence of these composite features allows the model to differentiate between teams with similar efficiency profiles but disparate levels of overall regular season dominance.

Model Implementation

The linear regression model is fit separately for the men's and women's divisions using Ordinary Least Squares (OLS). This foundational method identifies the specific coefficient vector $\hat{\boldsymbol{\beta}}$ that minimizes the sum of squared residuals between the predicted and observed scores across all training observations. The final curated men's training matrix consisted of $10,326$ rows and 6 features, while the women's training matrix comprised $10,828$ rows and 5 features. The formal specification for the men's model is defined as:

(16)

$$\begin{aligned}\text{Points}_A = \beta_0 &+ \beta_1(\texttt{OffRating}_A) + \beta_2(\texttt{DefRating}_B) + \beta_3(\texttt{AdjTempo}_A)\\ &+ \beta_4(\texttt{AdjTempo}_B) + \beta_5(\texttt{NetRtg}_A - \texttt{NetRtg}_B) + \beta_6(\texttt{Elo}_A - \texttt{Elo}_B) + \varepsilon\end{aligned}$$

and the corresponding specification for the women's model is:

$$\begin{aligned} \text{Points}_A = \beta_0 + \beta_1(\texttt{AdjOE}_A) + \beta_2(\texttt{AdjDE}_B) + \beta_3(\texttt{AdjTempo}_A) \\ + \beta_4(\texttt{AdjTempo}_B) + \beta_5(\texttt{Elo}_A - \texttt{Elo}_B) + \varepsilon \end{aligned} \tag{17}$$

where $\varepsilon \sim \mathcal{N}(0, \hat{\sigma}^2)$ is the residual error term and $\hat{\sigma}$ is estimated directly from the training data. Note that $(\texttt{NetRtg}_A - \texttt{NetRtg}_B)$ and $(\texttt{Elo}_A - \texttt{Elo}_B)$ correspond directly to `Diff_NetRtg` and `Diff_Elo` in the model implementation, respectively, representing the net rating differential and Elo rating differential between the two competing teams.

The decision to train entirely separate models for each division directly mirrors the justification applied in the earlier logistic regression section. The fundamental relationship between efficiency ratings, tempo, and scoring may differ structurally between divisions due to inherent differences in average pace, physicality, and overall playing style. Forcing shared parameter estimates across both genders would carry a significant risk of masking these meaningful structural differences, thereby degrading predictive accuracy. The resulting values of $\hat{\sigma}$ for both divisions are explicitly reported and interpreted in the Model Analysis section, where they serve as a direct measure of model calibration and as the primary input for the subsequent simulation stage.

To convert the predicted point totals into win probabilities, a Monte Carlo simulation framework is applied to each specific tournament matchup. For a given matchup between Team A and Team B, the fitted linear regression model produces two distinct point estimates, denoted as $\hat{\mu}_A$ and $\hat{\mu}_B$, representing the statistically expected scores for each team given their respective efficiency profiles. Continuous score realizations are then drawn independently for each team from a normal distribution centered on these predictions:

$$\text{Points}_A \sim \mathcal{N}(\hat{\mu}_A, \hat{\sigma}^2) \tag{18}$$

$$\text{Points}_B \sim \mathcal{N}(\hat{\mu}_B, \hat{\sigma}^2) \tag{19}$$

Following these simulated iterations, the overarching win probability for Team A is computed as the strict proportion of trials where their simulated score exceeds that of their opponent:

$$\hat{P}(\text{Team A wins}) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}[\text{Points}_{A,i} > \text{Points}_{B,i}] \tag{20}$$

This final calculated proportion is submitted directly to Kaggle as the predicted win probability for the specified matchup, with the complementary probability $1 - \hat{P}(\text{Team A wins})$ assigned to Team B. The normality assumption underlying this simulation stage is robustly supported by both theoretical foundations and empirical evidence. From a theoretical perspective, a team's final score is essentially the aggregate sum of many independent possession outcomes over the course of a game. By the Central Limit Theorem [22], the distribution of this cumulative aggregate naturally converges toward a normal distribution as the total number of possessions increases. To verify this phenomenon empirically, the full distribution of regular season scores across all available historical seasons was formally examined for both divisions. As shown visually in Figures 12 and 13, the observed score distributions closely trace a fitted normal curve in both datasets.

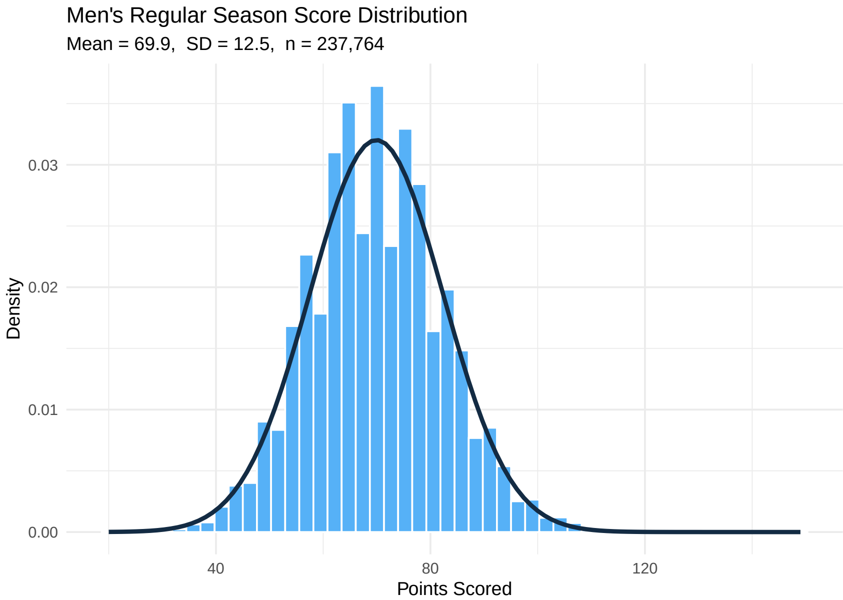

Figure 12. Distribution of Men's Regular Season Scores

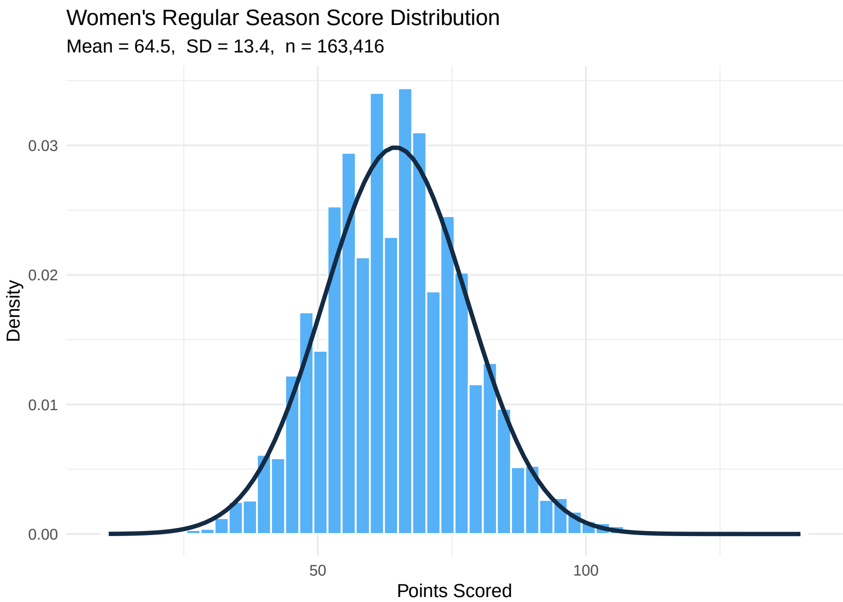

Figure 13. Distribution of Women's Regular Season Scores

During the final prediction phase, the men's model utilized $n = 100,000$ simulation iterations per matchup, while the women's 2025 submission utilized $n = 50,000$. Fortunately, at both of these massive sample sizes, the standard error of the resulting proportion estimate is strictly bounded by $\sqrt{p(1-p)/n} \leq 0.002$ in the absolute worst case scenario at $n = 50,000$. This mathematical bound guarantees that the simulated probability estimates have successfully converged to highly stable values that are immune to random sampling noise. Ultimately, the predictions for the 2025 tournament are generated by computing the expected scores for every possible matchup using each team's 2025 KenPom and Bart Torvik efficiency ratings alongside their end-of-season Elo ratings, and then systematically applying this Monte Carlo simulation to produce a final win probability $\hat{P} \in (0, 1)$ for each unique pairing.

Model Analysis

With the models estimated on the 2025 regular season data, the analysis begins by verifying that the four core assumptions of multiple linear regression are satisfied before interpreting the coefficient estimates. These assumptions are linearity, normality of residuals, homoscedasticity.

These three assumptions are assessed jointly through the combined diagnostic plots presented in Figures 14 and 15. The first assumption, linearity, requires that the relationship between the dependent variable and each independent variable is linear. The Residuals vs. Fitted panels show approximately flat smoothing lines across the range of predicted scores, confirming that the linearity assumption is reasonably satisfied in both models, though some variation in residual spread is visible. The second assumption, normality of residuals, requires that the residuals follow an approximately normal distribution. The Q-Q panels show that the residuals follow the theoretical normal reference line closely through the central range in both models, with some deviation in the tails driven by a small number of extreme blowout games. The third assumption, homoscedasticity, requires that the variance of the residuals remains consistent across all levels of the predicted values. The variation in residual spread noted in the Residuals vs. Fitted panels suggests the homoscedasticity assumption may not be fully satisfied. The Breusch-Pagan test returned a statistically significant result. However, classical heteroscedasticity tests are known to be sensitive to outliers and high leverage points, which can inflate the rejection rate well beyond the nominal level [29]. The Residuals vs. Leverage panel confirms the presence of high leverage points in both models, suggesting the significant Breusch-Pagan result should be interpreted with caution. As the OLS coefficient estimates remain unbiased under heteroscedasticity, and the model is used for score prediction rather than formal inference, we proceed with the analysis while acknowledging non-constant variance as a limitation.

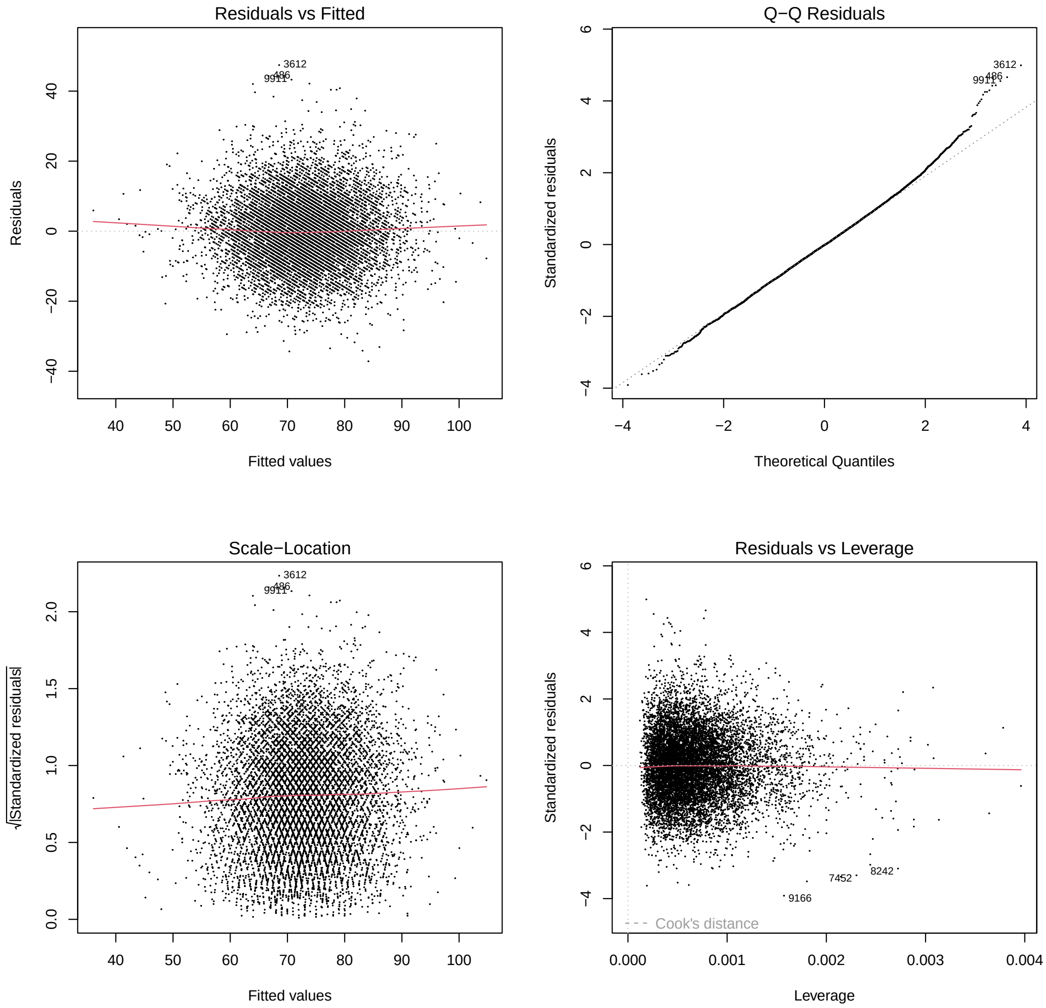

Figure 14. Diagnostic Plots for the Men's Linear Regression Model.

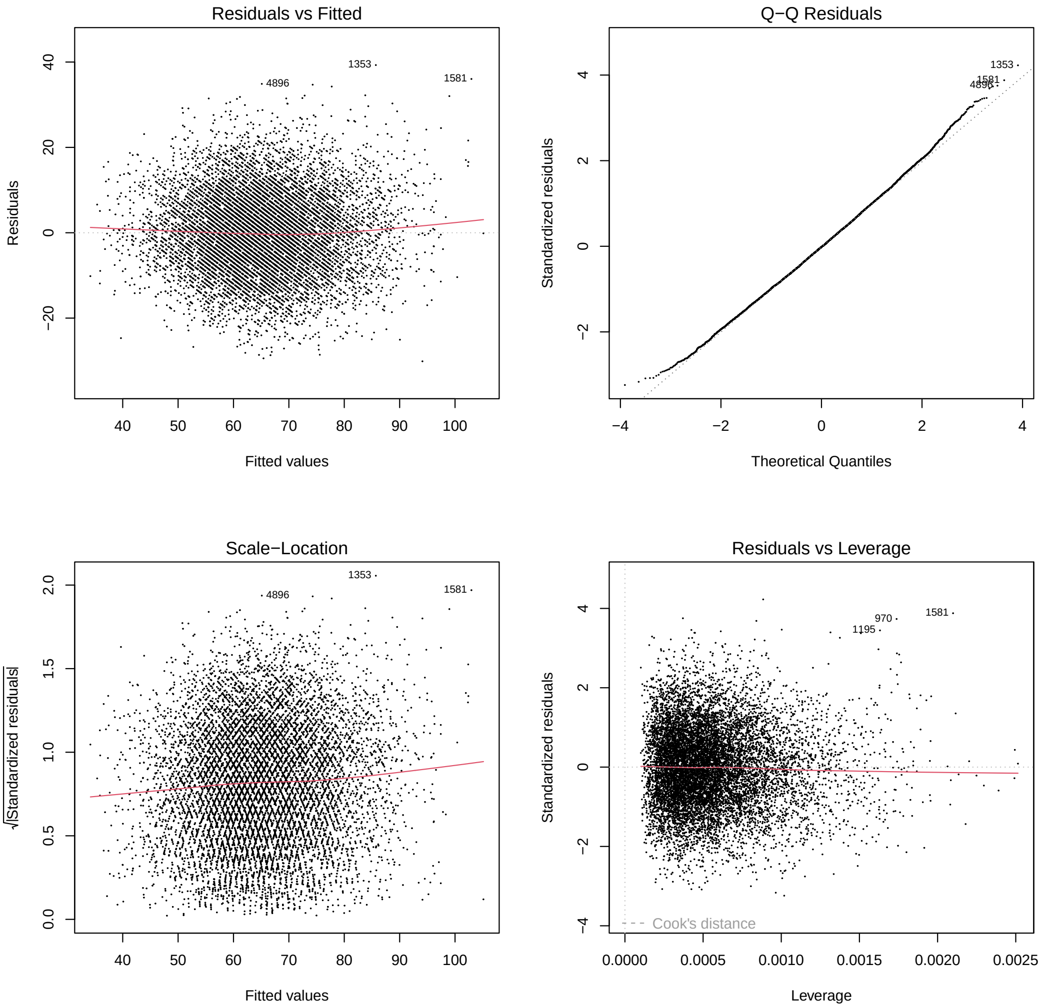

Figure 15. Diagnostic Plots for the Women's Linear Regression Model.

Lastly, we check the model for multicollinearity. Multicollinearity requires that the independent variables are not too highly correlated with one another. Figure 16 presents the VIF plots for both the men's and women's Monte Carlo linear regression models. In the men's model, `Diff_NetRtg` carries a moderate VIF of approximately 7.5, the highest among all predictors. While this falls below the problematic threshold of 10, it indicates meaningful shared variance between Diff_NetRtg and Diff_Elo, as both features capture overall team quality. In the women's model, all VIF values fall below 5, indicating a clean feature structure with no problematic collinearity.

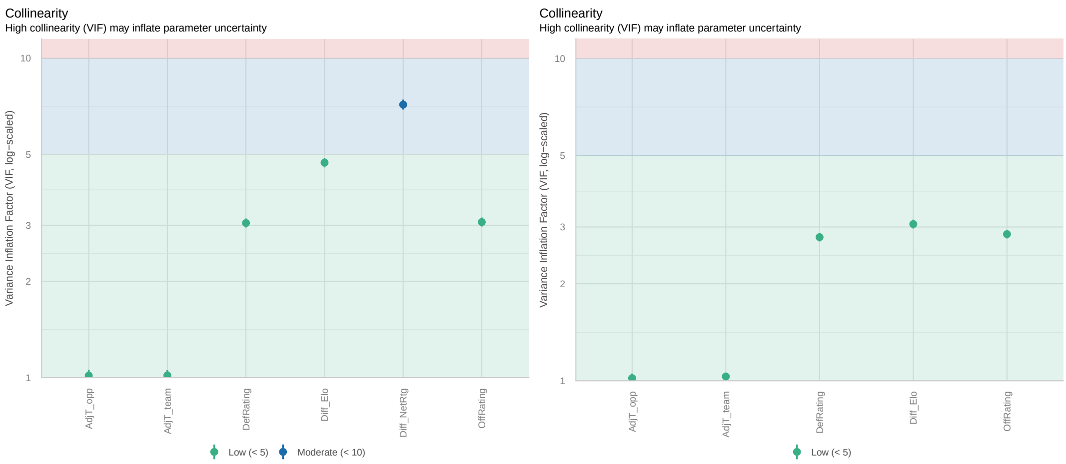

Figure 16. VIF Plots for the Men's (left) and Women's (right) Monte Carlo Linear Regression Models.

The pairwise correlation matrices in Figure 17 confirm these findings. In the men's model, $(\texttt{NetRtg}_A - \texttt{NetRtg}_B)$ and $(\texttt{Elo}_A - \texttt{Elo}_B)$ exhibit a correlation of 0.89, the strongest relationship in the matrix and the primary driver of the elevated VIF for $(\texttt{NetRtg}_A - \texttt{NetRtg}_B)$. All other pairwise correlations in the men's model remain below 0.5. In the women's model, with $(\texttt{NetRtg}_A - \texttt{NetRtg}_B)$ absent, the strongest relationship is between AdjOE and $(\texttt{Elo}_A - \texttt{Elo}_B)$ at 0.46, which is well within acceptable bounds. A moderate negative correlation of -0.35 between OffRating and DefRating is also present in the men's model, with a similar value of -0.37 observed in the women's model. This is expected given that teams facing stronger defenses tend to have lower offensive outputs, meaning that higher defensive ratings naturally correspond to lower scoring environments for the opposing offense.

While the men's model's moderate collinearity does not invalidate the coefficient estimates, it inflates the standard errors of the affected predictors and makes it more difficult to isolate their individual contributions. Both features were retained for their theoretical relevance rather than purely statistical grounds.

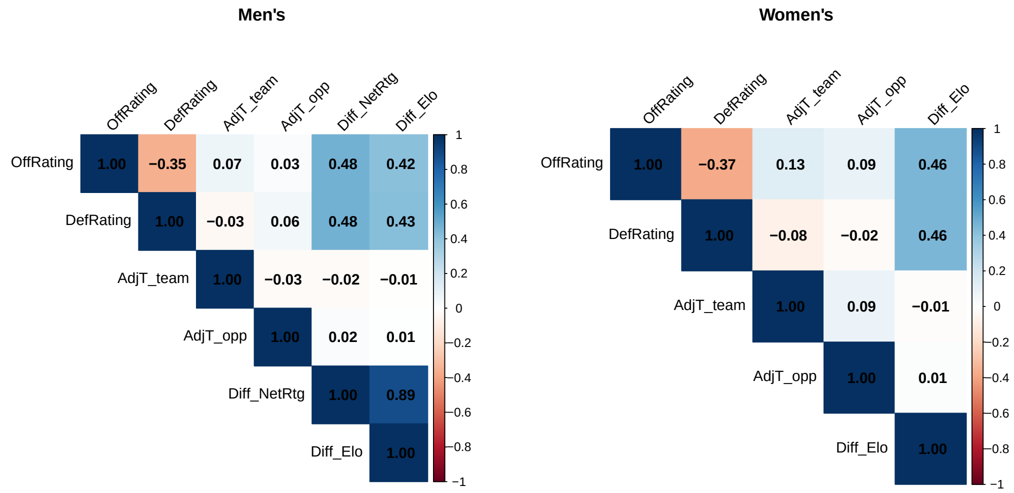

Figure 17. Feature Correlation Matrices for the Men's and Women's Linear Regression Models.

Because separate models were trained for the men's and women's tournaments, the coefficient structures can be analyzed independently and then compared to assess whether the efficiency and tempo features operate differently across divisions. Tables 7 and 8 present the estimated coefficients, standard errors, p-values, and significance levels for the men's and women's models respectively.

In the men's model, four of the six predictors reach statistical significance at the 0.001 level: `OffRating`, `DefRating`, `AdjT_team`, and

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	-207.3986	5.4365	0.0000	***
Off Rating	0.6519	0.0216	0.0000	***
Def Rating	0.6421	0.0239	0.0000	***
AdjT Team	1.0826	0.0378	0.0000	***
AdjT Opp	1.0086	0.0378	0.0000	***
Diff NetRtg	0.0540	0.0181	0.0029	**
Diff Elo	0.0013	0.0008	0.1229	

Table 7. Linear Regression Coefficient Estimates (Men's 2025 Model).

Variable	Estimate	Std. Error	p-value	Sig.
Intercept	-202.9007	3.9728	0.0000	***
Off Rating	0.7309	0.0136	0.0000	***
Def Rating	0.7866	0.0189	0.0000	***
AdjT Team	0.9605	0.0333	0.0000	***
AdjT Opp	0.8542	0.0331	0.0000	***
Diff Elo	0.0024	0.0006	0.0000	***

Table 8. Linear Regression Coefficient Estimates (Women's 2025 Model).

`AdjT_opp`. The positive coefficient on `OffRating` ($\hat{\beta} = 0.6519$) confirms that teams with higher adjusted offensive efficiency are expected to score more points, consistent with the feature's definition. The positive coefficient on `DefRating` ($\hat{\beta} = 0.6421$) reflects that facing a weaker defense is associated with higher expected scoring. Both tempo features carry positive and similarly sized coefficients, with `AdjT_team` ($\hat{\beta} = 1.083$) and `AdjT_opp` ($\hat{\beta} = 1.009$) indicating that faster pace from either team is associated with more points scored. `Diff_NetRtg` reaches significance at the 0.01 level ($p = 0.0029$), contributing a small but meaningful positive adjustment to the expected score when a team's net rating differential is favorable. `Diff_Elo` fails to reach significance ($p = 0.1229$), which is directly attributable to its collinearity with `Diff_NetRtg` as identified in the VIF analysis.

The women's model tells a somewhat different story. All five predictors reach statistical significance, including Diff_Elo ($\hat{\beta} = 0.002383$, $p < 0.001$), which was insignificant in the men's model. The efficiency coefficients are larger in magnitude for the women's model than the men's, with OffRating at 0.7309 versus 0.6519 and DefRating at 0.7866 versus 0.6421, suggesting that adjusted efficiency ratings are more predictive of actual scoring in the women's game. The tempo coefficients in the women's model are slightly smaller than the men's, with AdjT_team at 0.9605 versus 1.083 and AdjT_opp at 0.8542 versus 1.009, suggesting that pace has a somewhat weaker impact on scoring in the women's game. The significance of Diff_Elo in the women's model but not the men's is directly attributable to the absence of Diff_NetRtg in the women's feature set. With Diff_NetRtg absent, Diff_Elo becomes the primary carrier of the team quality differential signal and its contribution is clearly detectable.

The two models also differ meaningfully in their goodness of fit. The women's model achieves an R^2 of 0.524, meaning it explains 52.4% of the variance in points scored, compared to 0.379 for the men's model. This gap suggests that Bart Torvik efficiency ratings are more predictive of women's scoring outcomes than KenPom ratings are for men's, possibly because the women's game exhibits less of the high variance play that characterizes men's basketball at the tournament level [9]. The residual standard deviations are $\hat{\sigma} = 9.507$ for the men's model and $\hat{\sigma} = 9.300$ for the women's model, meaning the women's model's score predictions are marginally more precise. These values feed directly into the Monte Carlo simulation as the spread of the score distribution, so the women's slightly lower $\hat{\sigma}$ means its win probability estimates are derived from slightly tighter score distributions.

Taken together, the diagnostic analysis confirms that both models satisfy the core assumptions of linear regression sufficiently to justify their use in the simulation framework. The women's model is the stronger of the two by every measured criterion, achieving higher R^2, lower $\hat{\sigma}$, cleaner VIF structure, and full predictor significance. The men's model is adequate but limited by the moderate collinearity between `Diff_NetRtg` and `Diff_Elo` and the lower overall explanatory power of the efficiency features in the men's regular season data. These findings reinforce the decision to train the models independently rather than pooling the divisions, as the structural differences in coefficient magnitudes, predictor significance, and goodness of fit confirm that the two divisions are not governed by the same scoring dynamics.

RESULTS

The Results section evaluates the predictive performance of all three models developed in this analysis on the 2025 NCAA tournament. The actual tournament outcomes serve as the ground truth against which all predictions are evaluated. Performance is assessed across three complementary dimensions: classification accuracy derived from the confusion matrices, the AUC curves computed on the 2025 tournament predictions, and the combined Brier Score across both divisions. Together these metrics capture both the directional accuracy of each model's winner predictions and the quality of its probability calibration. The Kaggle competition winning Brier Score of 0.10411 serves as the external benchmark throughout.

2025 Results

The 2025 NCAA tournament serves as the primary evaluation for all three models developed in this paper. Each model generated win probability estimates for every possible tournament matchup across both the men's and women's divisions, and those probabilities are evaluated here against the actual tournament outcomes. Performance is assessed across three dimensions: classification accuracy derived from the confusion matrices, the Area Under the Receiver Operating Characteristic Curve computed on the 2025 tournament predictions, and the combined Brier Score across both divisions. Together these metrics provide a comprehensive picture of each model's predictive ability, capturing both directional accuracy and probability calibration. The Kaggle competition winning Brier Score of 0.10411 serves as the external benchmark against which all three models are compared.

The logistic regression model serves as the interpretable baseline and produces the most modest classification performance of the three models. Figure 18 presents the confusion matrices for the men's, women's, and combined results. The women's model achieves the strongest classification performance at 77.6% accuracy, followed by the combined model at 73.9% and the men's model at 70.1%. This ordering is consistent with the coefficient analysis in the Model Analysis section, where the women's model benefited from three statistically significant predictors while the men's model had only one. All three accuracy figures comfortably exceed the 50% baseline expected from random prediction, confirming that the Four Factors carry prediction power in both divisions.

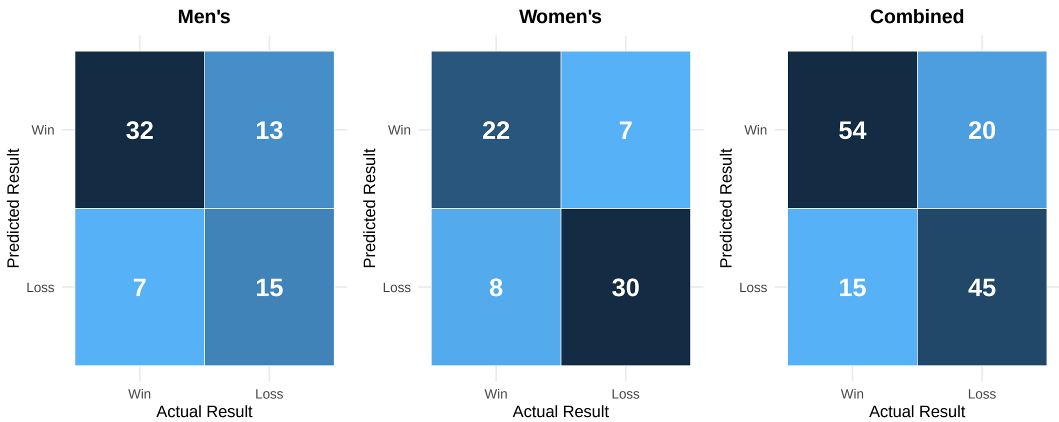

Figure 18. Confusion Matrices for the 2025 Logistic Regression Model

The XGBoost model produces stronger classification performance across both divisions, reflecting the expanded feature set and nonlinear modeling framework. Figure 19 presents the confusion matrices for the men's, women's, and combined results. The women's model achieves the strongest classification performance at 80.6% accuracy, followed by the combined model at 79.1% and the men's model at 77.6%. This ordering is consistent with the feature importance analysis, where measures of team quality such as seeding and Elo drive predictions, and the women's tournament historically exhibits stronger seed correlation with outcomes than the men's. All three accuracy figures improve upon the logistic regression results, confirming that the richer feature set contributes meaningful predictive gains over the Four Factors baseline.

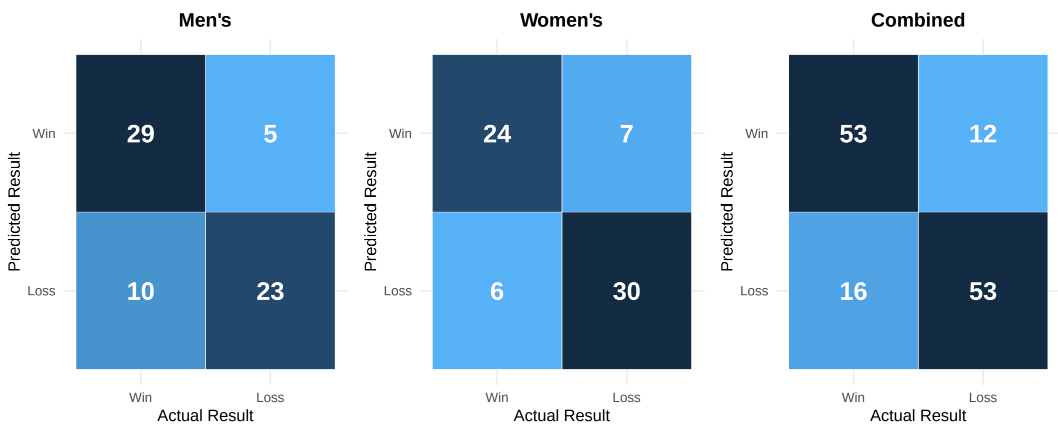

Figure 19. Confusion Matrices for the 2025 XGBoost Model.

The Linear Regression Monte Carlo model produces the strongest classification performance of the three models across both divisions. Figure 20 presents the mens,women's, and combined results. The women's model achieves 82.09% accuracy and the men's model achieves 77.61%, yielding a combined accuracy of 79.85%. The women's accuracy of 82.09% is the highest recorded across all three models and both divisions, consistent with the Model Analysis finding that the women's linear regression achieved a higher R^2 and lower $\hat{\sigma}$ than the men's model. The men's accuracy of 77.61% is comparable to the XGBoost men's result of 77.6%, suggesting that despite using a fundamentally different prediction approach, the two models identify similar matchup patterns in the men's tournament.

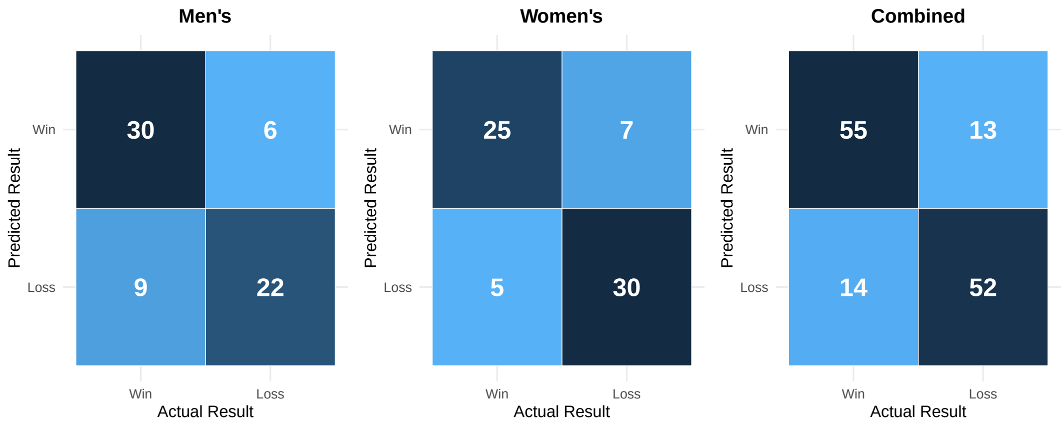

Figure 20. Confusion Matrices for the 2025 Monte Carlo Model.

Figure 21 presents the ROC curves for all three models evaluated on the 2025 tournament predictions. The ROC curve plots the true positive rate against the false positive rate across all possible classification thresholds, with the area under the curve summarizing overall predictive ability. A model with no predictive power would follow the diagonal reference line with an AUC of 0.5, while a perfect model would achieve an AUC of 1.0.

While the AUC results highlight the discriminative advantage of XGBoost and Monte Carlo over the logistic regression baseline, the Brier Scores tell a consistent story. Summarized in Table 9, the logistic regression achieves a combined Brier Score of 0.18401, while XGBoost achieves 0.12761 and the Monte Carlo model achieves 0.12857. To contextualize these values, a Brier Score of B implies that the model performed on average similarly to consistently assigning a win probability of $1 - \sqrt{B}$ to the eventual winner. By this measure the logistic regression corresponds to an average assigned probability of approximately 0.571, while XGBoost and Monte Carlo both correspond to approximately 0.643 and 0.641 respectively, reflecting meaningfully sharper probability estimates.

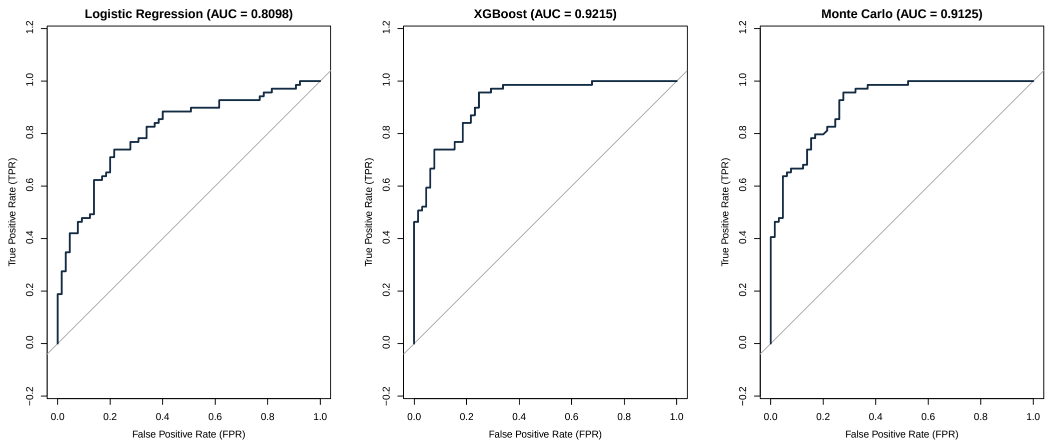

Figure 21. ROC Curves for All Three Models on the 2025 Tournament

The competition winning score of 0.10411 corresponds to approximately 0.677, indicating that top submissions are consistently more confident on clear matchups while remaining appropriately uncertain on competitive ones. The difference of less than 0.001 between XGBoost and Monte Carlo is negligible in practical terms, confirming that both models produce comparably calibrated win probabilities on the 2025 tournament despite their fundamentally different approaches. All three models fall short of the competition winning Brier Score of 0.10411, indicating that the best performing submissions in the Kaggle competition produce sharper and better calibrated probabilities than any of the three models developed here.

Given the comparable performance of the XGBoost and Monte Carlo models across all three evaluation metrics, the selection of a model for the

Model	Men's Acc	Women's Acc	Combined Acc	Brier Score	AUC
Logistic Regression	70.10%	77.60%	73.90%	0.18401	0.8098
XGBoost	77.60%	80.60%	79.10%	0.12761	0.9215
Monte Carlo	77.61%	82.09%	79.85%	0.12857	0.9125
Kaggle Winner	—	—	—	0.10411	—

Table 9. Model Performance Comparison on the 2025 Tournament

2026 Kaggle submission was not driven by marginal differences in Brier Score or AUC. The Monte Carlo model was selected for submission primarily on the basis of novelty. The simulation based scoring framework, which grounds win probability estimates in explicit score predictions derived from external efficiency ratings rather than learning a direct mapping from features to outcomes, represents a fundamentally different approach to the tournament prediction problem than the ensemble methods that dominate top Kaggle submission. From an academic standpoint the Monte Carlo model also offers greater interpretability, the predicted scores, the residual standard deviation $\hat{\sigma}$, and the simulated score distributions all carry direct basketball meaning that can be examined and explained in ways that the internal structure of a boosted tree ensemble cannot. The 2025 results confirm that this interpretability does not come at a meaningful cost to predictive performance, as the Monte Carlo model achieved the highest combined classification accuracy of the three models and a Brier Score within 0.001 of XGBoost.

2026 Retraining

The Linear Regression Monte Carlo model was retrained on the 2026 regular season data prior to submission, following the same pipeline described in the Methods section. The men's model was trained on $11,236$ observations across six features and the women's model was trained on 10,900 observations across five features, both reflecting a full 2026 regular season. The linearity, normality, and homoscedasticity assumptions were verified through diagnostic plots and were consistent with the 2025 findings. The collinearity structure, however, showed a notable change in the men's model. The VIF analysis, presented in Figure 22, reveals that both `Diff_NetRtg` and `Diff_Elo` now register in the moderate range, with `Diff_NetRtg` at approximately 8.0 and `Diff_Elo` at approximately 5.5, compared to 7.5 and below 5 respectively in 2025. Both values remain below the problematic threshold of 10, so the coefficient estimates remain valid, but the increased collinearity between these two features is a known limitation of the 2026 men's model. The women's model VIF structure remained clean with all values below 5, consistent with 2025.

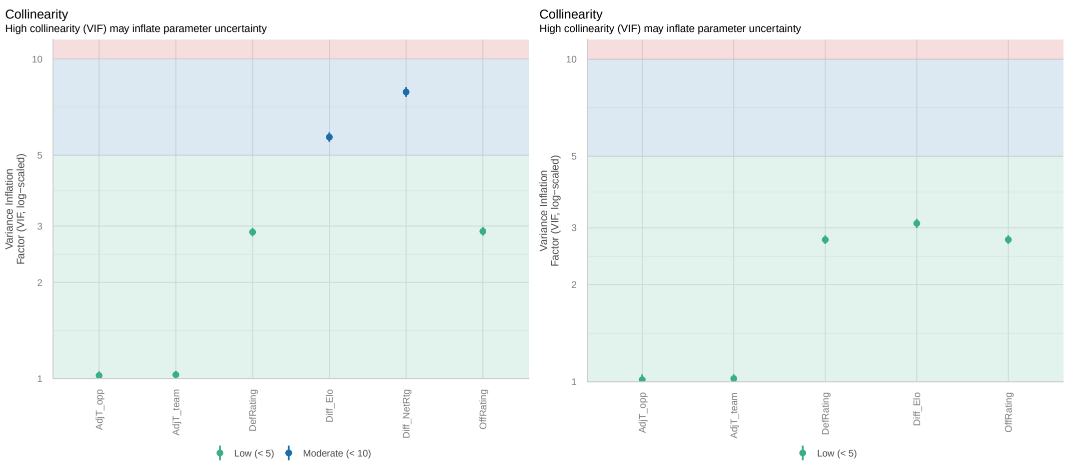

Figure 22. VIF Plots for the Men's and Women's 2026 Linear Regression Models

The 2026 men's model produced several notable differences from the 2025 version. Most significantly, `Diff_Elo` reached statistical significance at the 0.001 level ($p = 0.000101$) with a negative coefficient of -0.003569, reversing the positive but insignificant direction observed in 2025. This suggests that in the 2026 season, after controlling for KenPom efficiency ratings and net rating differential, teams with historically higher Elo ratings relative to their opponents scored slightly fewer points than expected, a possible reflection of season specific regression toward the mean for historically dominant programs. `Diff_NetRtg` also strengthened considerably, with its coefficient rising from 0.05398 in 2025 to 0.1284 in 2026 and reaching significance at the 0.001 level. The R^2 improved modestly to 0.4021 from 0.379, and σ increased slightly to 9.676 from 9.507.

The 2026 women's model remained structurally consistent with its 2025 counterpart. All five predictors reached statistical significance, with `OffRating` ($\hat{\beta} = 0.7369$), `DefRating` ($\hat{\beta} = 0.7750$), `AdjT_team` ($\hat{\beta} = 0.9290$), `AdjT_opp` ($\hat{\beta} = 0.8534$), and `Diff_Elo` ($\hat{\beta} = 0.002479$, $p < 0.001$) all performing consistently with their 2025 estimates. The efficiency coefficients are nearly identical to the 2025 values of 0.7309 and 0.7866 respectively, suggesting that the relationship between Bart Torvik efficiency ratings and scoring is stable across seasons. The R^2 improved slightly to 0.531 from 0.524, and σ increased marginally to 9.413 from 9.300, indicating comparable predictive precision to the prior year. The stability of the women's model across both seasons reinforces the reliability of the Bart Torvik efficiency framework as a consistent predictor of women's tournament scoring.

Variable	Estimate	Std. Error	p-value	Sig.
(Intercept)	-219.1897	5.1350	0.0000	***
OffRating	0.6784	0.0191	0.0000	***
DefRating	0.6744	0.0224	0.0000	***
AdjT_team	1.0885	0.0402	0.0000	***
AdjT_opp	1.0821	0.0400	0.0000	***
Diff_NetRtg	0.1284	0.0182	0.0000	***
Diff_Elo	-0.0036	0.0009	0.0001	***

Table 10. Linear Regression Coefficient Estimates (Men's 2026 Model).

Variable	Estimate	Std. Error	p-value	Sig.
(Intercept)	-197.5640	3.9846	0.0000	***
OffRating	0.7369	0.0131	0.0000	***
DefRating	0.7750	0.0183	0.0000	***
AdjT_team	0.9290	0.0325	0.0000	***
AdjT_opp	0.8534	0.0324	0.0000	***
Diff_Elo	0.0025	0.0006	0.0000	***

Table 11. Linear Regression Coefficient Estimates (Women's 2026 Model).

2026 Results

Prior to submitting the 2026 predictions, a set of manual probability overrides were applied to specific matchups based on historical seeding patterns in the NCAA tournament. These overrides, 32 in total, reflect the well documented dominance of top seeds in early rounds and were applied independently for the men's and women's divisions. The decision to implement deterministic overrides rather than relying purely on the model's calibrated probabilities was driven by the competitive nature of the Kaggle tournament. Securing a top leaderboard position (a "medal") requires differentiating the submission from the consensus, which often necessitates an aggressive strategy. While a precise model might accurately assign a 0.95 win probability to a 1-seed, that remaining 0.05 uncertainty still accumulates a small Brier Score penalty even when the favored team wins. By overriding these highly predictable early round matchups to a full 1.0 probability, the submission intentionally accepts the catastrophic risk of a historic upset in exchange for accumulating exactly zero penalty on expected outcomes, effectively maximizing the ceiling of the final score.

For the men's tournament, all 1 and 2 seeds were assigned a win probability of 1.0 in Round 1, reflecting the historical rarity of a 1 or 2 seed losing to a 15 or 16 seed opponent. In Round 2, all 1 seeds were assigned a win probability of 1.0, reflecting their historically strong performance against opponents in the round of 32. The women's overrides were more aggressive, consistent with the historically stronger seed correlation in the women's tournament. All 1 and 2 seeds were assigned a win probability of 1.0 in Round 1, and all 1 and 2 seeds were again assigned 1.0 in Round 2. In Round 3, all 1 seeds were assigned a win probability of 1.0. Finally, the UConn women's program was assigned a win probability of 1.0 in Round 4, reflecting their sustained dominance in the women's tournament, and because they entered the tournament undefeated. These overrides were applied after the Monte Carlo simulation produced its base probability estimates, replacing the model's predicted probabilities for the specified matchups with deterministic assignments. All other matchup probabilities were left unchanged from the simulation output.

Of the 32 matchups where overrides were applied, 30 resulted in the correct outcome, meaning the team assigned a probability of 1.0 did in fact win. The two incorrect overrides assigned a probability of 1.0 to the eventual loser, generating the maximum possible squared error of 1.0 per game. This is the inherent risk of a deterministic strategy. The final submitted Brier Score with overrides was 0.13237, compared to 0.12797 for the base Monte Carlo model without overrides. While the 30 correct overrides each contributed zero squared error and improved the Brier Score relative to what the base model would have assigned to those games, the two incorrect overrides each contributed a squared error of 1.0, the maximum possible penalty. The net effect of all 32 overrides combined was an inflation of the Brier Score by approximately 0.00440, meaning the damage from the two incorrect overrides outweighed the cumulative benefit of the 30 correct ones. This finding highlights a fundamental tension in tournament prediction between exploiting high confidence prior knowledge about seeding dominance and preserving the calibration properties of a probabilistic model. The net effect of a perfect override strategy would therefore have been a meaningfully lower Brier Score than either submission achieved.

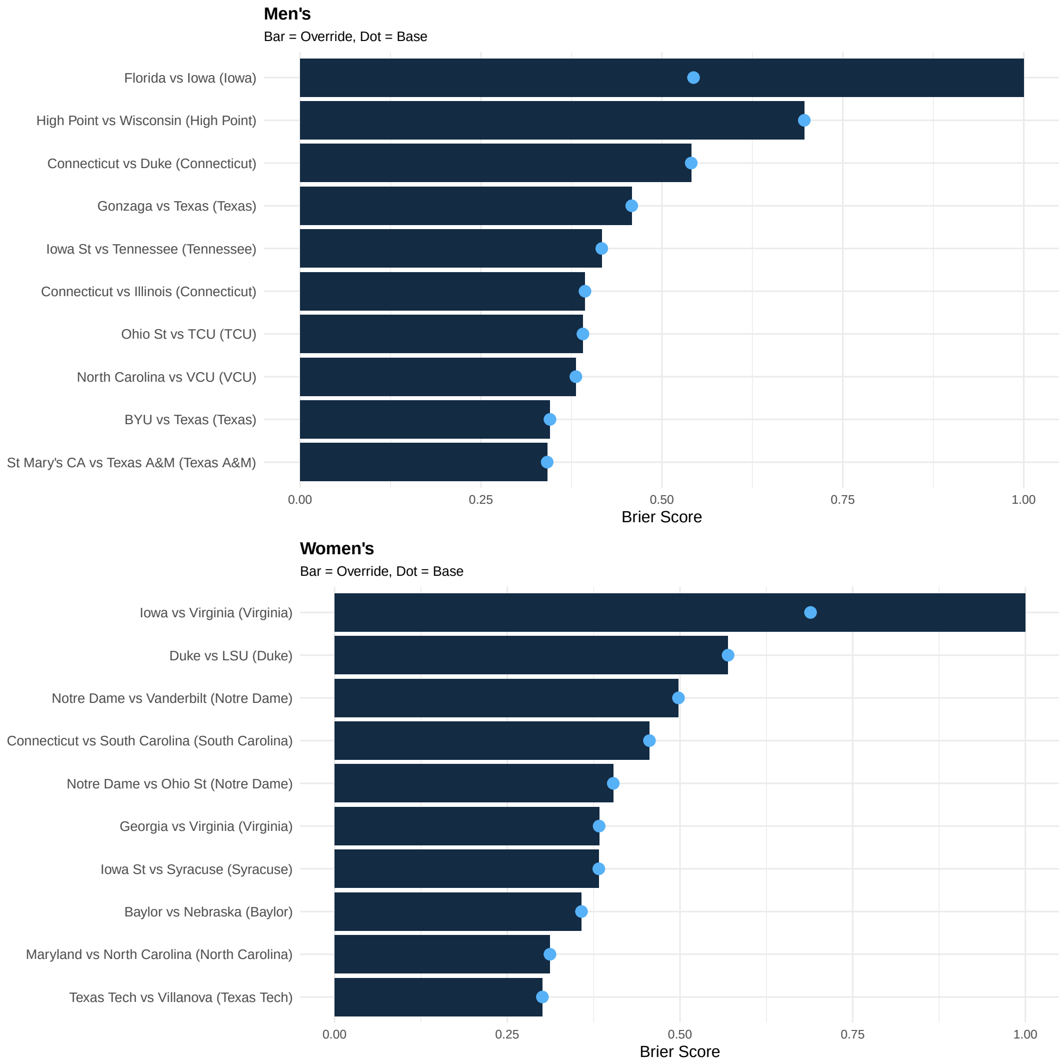

Figure 23. Top 10 Worst Predicted Games by Division (2026)

The confusion matrices for the override and base submissions are identical since no override changed the predicted winner, only the confidence assigned to each outcome. Figure 24 presents the confusion matrices for the men's, women's, and combined results. The men's model achieves a classification accuracy of 74.60%, the women's model achieves 80.95%, and the combined accuracy is 77.78%. The women's model again outperforms the men's in directional accuracy, consistent with the pattern observed in the 2025 results and the stronger R^2 and lower σ of the women's linear regression. Both divisions exceed the 50% random baseline comfortably, confirming that the Monte Carlo model retains meaningful predictive power on the 2026 tournament.

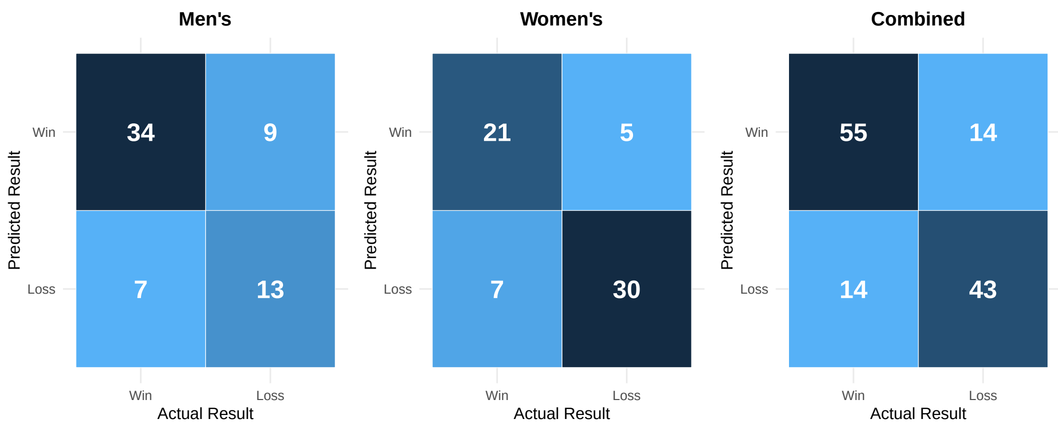

Figure 24. Confusion Matrices for the 2026 Monte Carlo Model

While the confusion matrices and classification accuracy are identical across both submissions, the ROC curves reveal meaningful differences in probability calibration. Figure 25 presents the ROC curves for both the override and base submissions. The base model achieves a combined AUC of 0.9021, outperforming the override submission's combined AUC of 0.8871. This gap is consistent with the Brier Score findings that the two incorrect overrides severely distorted the probability ranking that AUC measures. The base model also outperforms the override across both divisions individually, achieving an AUC of 0.8614 for men's and 0.9306 for women's, compared to 0.8370 and 0.9204 for the override submission respectively. The women's AUC of 0.9306 for the base model is the strongest discriminative performance recorded across any division in either submission, reinforcing the consistently superior performance of the women's model throughout this analysis.

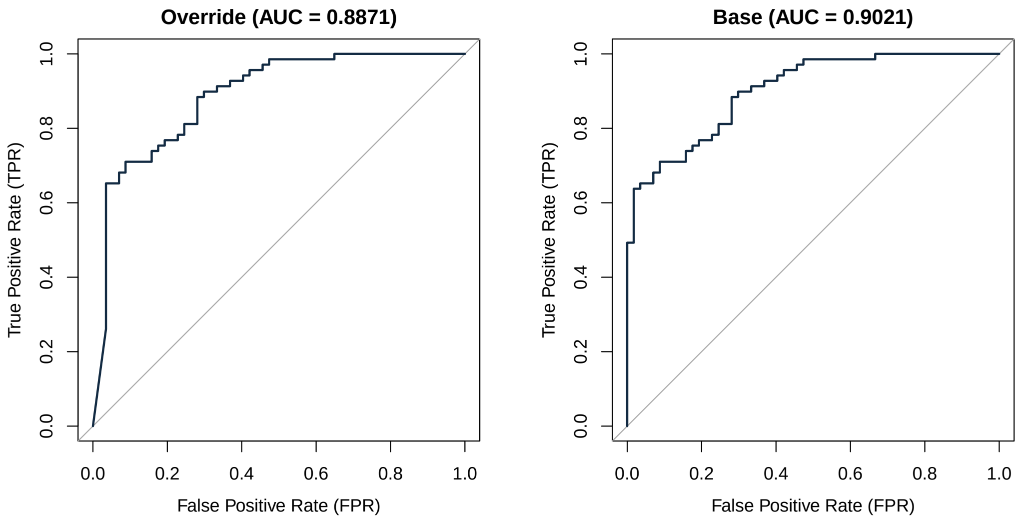

Figure 25. ROC Curves for the 2026 Monte Carlo Model

The 2026 Kaggle competition results provide important competitive context for evaluating both submissions. The base Monte Carlo model achieved a Brier Score of 0.12797, finishing 387th out of all submissions. The override submission achieved a Brier Score of 0.13237, finishing 822nd. The 8th place cutoff, the lowest medal position, was 0.11845, meaning the base model fell short by only 0.00352. The competition winning score was 0.10975. To contextualize these values, a Brier Score of 0.11845 corresponds to a win probability of approximately 0.656 assigned to the correct outcome, compared to 0.642 for the base model. In addition to the flat probability submission, a bracket simulation was run using the stochastic prediction framework described in the Methods section, where each game is predicted and Elo ratings are updated in real time after each result. The bracket correctly predicted 26 of 32 first round games, 12 of 16 second round games, 5 of 8 Sweet Sixteen games, 2 of 4 Elite Eight games, and 1 of 2 Final Four games, and correctly identified Michigan and UCLA as the national champion for the men's and women's respectively.

CONCLUSION

This paper developed and evaluated three distinct predictive models for NCAA March Madness tournament outcomes, motivated by the Kaggle March Machine Learning Mania competition framework. The logistic regression model provided an interpretable baseline grounded in Dean Oliver's Four Factors framework, demonstrating that a parsimonious five feature model trained on a narrow two year window of tournament data can achieve meaningful predictive accuracy. The XGBoost model extended this framework considerably, incorporating 139 features across three tiers of increasing complexity and leveraging the full historical tournament record from 2003 to 2024 to produce the best probability calibration of the three models on the 2025 tournament. The Linear Regression Monte Carlo model offered a fundamentally different philosophical approach, decomposing the prediction problem into an explicit scoring stage and a simulation stage grounded in external efficiency ratings from KenPom and Bart Torvik.

Across all three evaluation dimensions on the 2025 tournament, the XGBoost and Monte Carlo models performed comparably and both represented substantial improvements over the logistic regression baseline. The XGBoost model achieved the best Brier Score of 0.12761 and the highest AUC of 0.9215, while the Monte Carlo model achieved the highest combined classification accuracy of 79.85% and a Brier Score of 0.12857 that was negligibly higher than XGBoost. The women's models consistently outperformed the men's models across all three approaches and all three metrics, a finding that held across both the 2025 and 2026 tournaments and is consistent with the historically stronger seed correlation in the women's tournament.

The Monte Carlo model was selected for the 2026 live competition submission on the basis of methodological novelty, as the simulation scoring framework represents a fundamentally different approach to tournament prediction than the ensemble methods that dominate top Kaggle submissions. A set of 32 deterministic probability overrides were applied to highly favored seeds in early rounds, of which 30 were correct. The two incorrect overrides inflated the final submitted Brier Score from 0.12797 to 0.13237. The base model finished 387th in the competition with a Brier Score of 0.10975 separating it from the winning submission. The bracket simulation correctly identified Michigan as the national champion, predicting 47 of 63 games correctly across all six rounds.

Several limitations of this analysis warrant acknowledgment. The three models are not directly comparable on equal footing due to meaningful differences in their training data and implementation. The logistic regression model trains on two years of tournament outcomes, the XGBoost model trains on all available tournament outcomes from 2003 to 2024, and the Monte Carlo model trains on a single regular season. These structural differences mean that performance gaps between models reflect not only differences in modeling approach but also differences in the volume and type of data each model was exposed to. A controlled comparison holding training data constant across all three models would be necessary to isolate the contribution of methodology from the contribution of data availability.

Beyond the comparability issue, the logistic regression model's two-year rolling training window constrains the amount of historical information available and may underfit structural patterns that persist across multiple seasons. The Monte Carlo model's decision to train exclusively on the current regular season introduces sensitivity to season specific variation that a multiyear strategy could mitigate. The absence of `Diff_NetRtg` from the women's Monte Carlo model represents a known asymmetry between the two divisions that future iterations should address.

Future work could explore several directions. Ensemble approaches that combine the probability estimates from multiple models may produce better outputs than any individual model. Incorporating injury reports, player availability data, or transfer portal activity could improve the model's sensitivity to mid season, and off season roster changes. A natural extension of the Monte Carlo framework would be to model individual player performance and aggregate those estimates to the team level, replacing the reliance on aggregate efficiency ratings with a scoring model that explicitly accounts for matchup player dynamics. This approach would be particularly valuable in the post-NIL era where roster composition changes rapidly. Finally, a systematic analysis of the optimal override threshold, the seed cutoff and round depth at which deterministic assignments improve rather than hurt the expected Brier Score, would provide a more principled foundation for the override strategy employed here.

REFERENCES

[1] Aster, R. C., Borchers, B., Thurber, C. H. (2018). *Parameter Estimation and Inverse Problems* (2nd edition). Academic Press.

[2] Berkowitz, S. “NCAA revenue shattered record in 2024, but association faces massive debt from legal settlements.” *USA Today.* (2025). Available at https://www.usatoday.com/story/sports/college/2025/02/27/ncaa-financial-report-revenue-surplus/80733616007/

[3] Boone, Kyle. “How NIL Has Changed College Basketball: Numbers Deep Dive Reveals Surprising Trends, Recipe for Success.” CBSSports.com, 2025, www.cbssports.com/college-basketball/news/how-nil-has-changed-college-basketball-numbers-deep-dive-reveals-surprising-trends-recipe-for-success/.

[4] “Bracketology Driven by Data — News Center.” Georgia Tech News Center, 2026, news.gatech.edu/news/2026/03/18/bracketology-driven-data.

[5] Brier, Glenn W. “Verification of Forecasts Expressed in Terms of Probability.” *Monthly Weather Review*, vol. 78, no. 1, Jan. 1950, pp. 1–3, https://doi.org/10.1175/1520-0493(1950)078%3C0001:vofeit%3E2.0.co;2.

[6] Burns, William. “Efficient-Market Hypothesis (EMH) — EBSCO.” EBSCO Information Services, Inc., 2024, www.ebsco.com/research-starters/social-sciences-and-humanities/efficient-market-hypothesis-emh.

[7] Cohen, Dara. “Americans to Legally Wager Estimated $3.1 Billion on March Madness - American Gaming Association.” American Gaming Association, 13 Mar. 2025, www.americangaming.org/americans-to-legally-wager-estimated-3-1-billion-on-march-madness/.

[8] Ferrer, Josep. “What Is Cross-Validation? A Plain English Guide with Diagrams - KDnuggets.” KDnuggets, 1 Oct. 2025, www.kdnuggets.com/what-is-cross-validation-a-plain-english-guide-with-diagrams.

[9] Figueroa, Jared. “Why Does the Women’s NCAA Tournament Feature so Few Upsets?” Cronkite News, 18 Mar. 2026, cronkitenews.azpbs.org/2026/03/18/women-ncaa-tournament-features-few-upsets/. Accessed 8 Apr. 2026.

[10] "Four Factors." Basketball-Reference.com, www.basketball-reference.com/about/factors.html.

[11] Glickman, Mark E., and Thomas Doan. *The US Chess Rating System*. 26 July 2025, www.glicko.net/ratings/rating.system.pdf.

[12] gotoConversion. "GitHub - GotoConversion/Goto_conversion: Powered $47,000 of Prize Money, over 10 Gold Medals and 100 Medals on Kaggle." GitHub, 2025, github.com/gotoConversion/goto_conversion/tree/main. Accessed 20 Jan. 2026.

[13] Gumm, J., A. Barrett, and G. Hu. "A Machine Learning Strategy for Predicting March Madness Winners." *2015 IEEE/ACIS 16th International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD)*, 2015, pp. 1–6, `https://doi.org/10.1109/SNPD.2015.7176206`.

[14] Harris, Janie. "How Hard Is It to Pick a Perfect Bracket? — NCAA.com." NCAA, 18 Feb. 2026, www.ncaa.com/news/basketball-men/bracketiq/2018-02-15/march-madness-bracket-challenge-how-hard-it-pick-perfect.

[15] Harville, David A. "The Selection or Seeding of College Basketball or Football Teams for Postseason Competition." *Journal of the American Statistical Association*, vol. 98, no. 461, March 2003, pp. 17–27, `https://doi.org/10.1198/016214503388619058`.

[16] Haupts, L. "Calculate Elo Ratings," Kaggle Notebook, 2025. [Online]. Available: `https://www.kaggle.com/code/lennarthaupts/calculate-elo-ratings`

[17] "Hoops in Motion: The Evolution of Pace in Basketball." Substack.com, SLT Sports Analytics, Mar. 2024, sltsportsanalytics.substack.com/p/hoops-in-motion-the-evolution-of.

[18] International Journal For Innovative Research In Multidisciplinary Field. "Jaro Winkler Fuzzy Match Algorithm to Calculate a Similarity Index between Two Strings Using Open Source Platform." Peer-Reviewed, Refereed, Indexed Journal with IC, no. 8, 2018, pp. 2455-0620, www.ijirmf.com/wp-content/uploads/IJIRMF201808019.pdf. Accessed 7 Apr. 2026.

[19] Kanade, Vijay. "Everything You Need to Know about Logistic Regression." Spiceworks Inc, 8 Apr. 2022, www.spiceworks.com/soft-tech/what-is-logistic-regression/.

[20] Kavlakoglu, Eda, and Erika Russi. "XGBoost." Ibm.com, 9 May 2024, www.ibm.com/think/topics/xgboost.

[21] Kizildemir, Melis, et al. "A Family of Solutions Related to Shin's Model for Probability Forecasts." *Journal of Quantitative Analysis in Sports*, vol. 21, no. 2, 6 Feb. 2025, pp. 153–158, https://doi.org/10.1515/jqas-2024-0064. Accessed 20 Jan. 2026.

[22] Kwak, Sang Gyu, and Jong Hae Kim. "Central Limit Theorem: The Cornerstone of Modern Statistics." *Korean Journal of Anesthesiology*, vol. 70, no. 2, 21 Feb. 2017, pp. 144–156, https://doi.org/10.4097/kjae.2017.70.2.144.

[23] "Law of Large Numbers — EBSCO." EBSCO Information Services, Inc., 2022, www.ebsco.com/research-starters/mathematics/law-large-numbers.

[24] Logistic Regression 12.1 Modeling Conditional Probabilities.

[25] Lumivero. "Ordinary Least Squares Regression (OLS)." XLSTAT, 2025, www.xlstat.com/solutions/features/ordinary-least-squares-regression-ols.

[26] Massey, Kenneth. *Massey Ratings*. masseyratings.com, 2025, https://masseyratings.com. Accessed Mar. 2025.

[27] mcubed.net. *Women's NCAA Basketball Tournament: Games between #2 seeds and #10 seeds*. Last updated April 6, 2026. Available at `http://mcubed.net/ncaabw/seed2v10.shtml`.

[28] Oliver, Dean. *Basketball on Paper*. Potomac Books, 1 Nov. 2004.

[29] Pechcińska, A. (2023). *The Robustness of Heteroscedasticity Tests*. Master's Thesis, Erasmus School of Economics, Erasmus University Rotterdam. Retrieved from `https://thesis.eur.nl/pub/70292/Thesis_AgnieszkaPechcinska_638592.pdf`

[30] Pomeroy, Ken. *Pomeroy College Basketball Ratings*. The Forecast Factory LLC, 2025, https://kenpom.com. Accessed Mar. 2025.

[31] raddar. "Kaggle March Machine Learning Mania 2024." *Kaggle*, 2024, `https://www.kaggle.com/competitions/march-machine-learning-mania-2024/discussion/494723`.

[32] Raschka, Sebastian. "LinearRegression: An Implementation of Ordinary Least-Squares Linear Regression - Mlxtend." Rasbt.github.io, rasbt.github.io/mlxtend/user_guide/regressor/LinearRegression/.

[33] Ruder, Sebastian. "An Overview of Gradient Descent Optimization Algorithms." *arXiv preprint arXiv:1609.04747*, 2017, https://arxiv.org/abs/1609.04747.

[34] Rudin, Cynthia. Cross Validation Duke Course Notes.

[35] Schwertman, Neil C., Kathryn L. Schenk, and Brett C. Holbrook. "More Probability Models for the NCAA Regional Basketball Tournaments." *The American Statistician*, vol. 50, no. 1, Feb. 1996, pp. 34–38, https://www.jstor.org/stable/2685041.

[36] Simon, Jay. "Inefficient Forecasts at the Sportsbook: An Analysis of Real-Time Betting Line Movement." *Management Science*, vol. 70, no. 12, Dec. 2024, pp. 8583–8611, https://doi.org/10.1287/mnsc.2022.00456. Accessed 6 Jan. 2025.

[37] Tavhare, Avadhoot. "Occam's Razor in Machine Learning: The Principle of Parsimony." Medium, 21 May 2024, medium.com/@qjbqvwzmg/occams-razor-in-machine-learning-the-principle-of-parsimony-de110ce7fe13.

[38] Torvik, Bart. "T-Rank FAQ." Blogspot.com, 2017, adamcwisports.blogspot.com/p/every-possession-counts.html.

[39] Yao, Xiaotong, Fu, Xiaoli, and Zong, Chaofei. (2022). Short-Term Load Forecasting Method Based on Feature Preference Strategy and LightGBM-XGboost. *IEEE Access*. 10.1109/ACCESS.2022.3192011.

[40] "XGBoost Regularization Techniques — XGBoosting." Xgboosting.com, 2024, xgboosting.com/xgboost-regularization-techniques/.

[41] Zimmermann, Albrecht, Sruthi Moorthy, and Zifan Shi. "Predicting College Basketball Match Outcomes Using Machine Learning Techniques: Some Results and Lessons Learned." *arXiv preprint arXiv:1310.3607*, 2013, https://arxiv.org/abs/1310.3607.